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AIR HANDLING UNIT DETAILED DESIGN

TITLE:

AIR HANDLING UNIT DETAILED CALCULATION FOR ACCOMODATION REPORT and DRAWING FOR MT. ANTARES 1

OWNER:



PROGRAM STUDI D4-TEKNIK PERMESINAN KAPAL
JURUSAN TEKNIK PERMESINAN KAPAL
POLITEKNIK PERKAPALAN NEGERI SURABAYA

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LOG SHEET

Air Conditioning & Ducting Design Project MT. ANTARES 1

NO .	ITEMS	OK .	RE V	CHECKED BY (DATE AND SIGN)
1	Cooling Load Calculation	√		
2	Duct Airflow Calculation	√		
3	Duct Material Selection	√		
4	Duct Sizing and Selection	√		
5	Insulation Material Selection and Thickness Determination	√		
6	Duct Routing`	√		
7	Ductwork Drawing (HVAC Duct Layout)	√		
8	Airflow Modeling and Simulation	√		
9	Refrigeration Equipment Layout and Plot Plan	√		
10	Refrigerant Selection	√		
11	Air Handling Unit (AHU) Specification and Selection	√		
12	Airflow Evaluation and Analysis	√		
13	Budget Estimation Plan			
14	Energy Efficiency Assessment			
15	Global Warming Potential (GWP) Assessment			
16	Economic Feasibility Analysis (or Life-Cycle Cost Analysis)			
17	Final Comprehensive Report			

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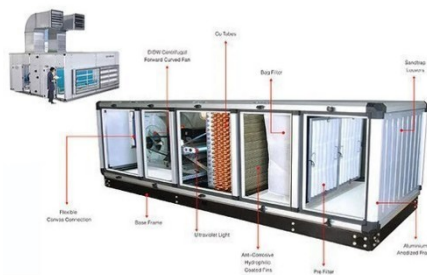
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1. INTRODUCTION

1.1. Project Overview

Kapal M.T. Antares I adalah kapal pengangkut produk minyak (oil products tanker) dengan panjang keseluruhan sekitar 179,88 meter dan lebar sekitar 32,20 meter, berbendera Liberia dan dibangun pada tahun 2008. Kapal ini memiliki daya angkut mati (deadweight) sekitar 45.997 ton serta tonase bruto terdaftar sebesar 28.049. Dalam proyek ini, fokus penelitian akan diarahkan pada sistem pengkondisian udara ruang akomodasi awak kapal, dengan mempertimbangkan kondisi lingkungan laut, beban termal dari peralatan mesin, serta syarat kenyamanan suhu dan kelembapan sesuai standar. Ruang akomodasi menjadi area penting karena awak kapal akan menempatnya dalam durasi panjang selama pelayaran, sehingga sistem HVAC (pemanasan, ventilasi, dan pendingin udara) harus dirancang agar efisien, andal, dan mampu menjaga kualitas udara ideal di tengah kondisi ekstrem di laut.

Air Handling Unit (AHU) adalah bagian dari system HVAC yang berguna untuk mengkondisikan dan mensirkulasi udara disuatu ruangan, tepatnya untuk MV ALAHAS. AHU dirancang dan dibuat khusus untuk pengkondisian udara luar agar sesuai dengan ruangan sehubungan dengan suhu, kelembaban dan kebersihan. AHU akan mengambil udara luar, mengkondisikan ulang dan meyalurkannya sebagai udara segar ke dalam ruangan, sehingga akan menghasilkan kualitas udara yang baik. Apabila suhu ruangan panas maka udara segar akan dipanaskan oleh unit pemulihan atau koil pemanas. Sedangkan apabila suhu ruangan dingin maka udara akan didinginkan oleh koil pendingin.



Gambar 1. 1 Gambar Air Handling Unit

Sumber : (air-handling-units-Indiamart.com)

1.2. Scope of Works

Tuliskan ruang lingkup pekerjaan di sini termasuk permasalahan, tujuan pekerjaan dst. Scope of work disusun berurutan sesuai urutan pekerjaan:

1. Pemilihan material dan ketebalan insulasi ruangan
2. Kalkulasi beban pada ruangan yang dilakukan pengkondisian udara.
3. Kalkulasi dan pemilihan material ducting dan perlengkapan saluran udara.
4. Kalkulasi dan pemilihan spesifikasi sistem air handling unit.

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1.3. Basic Theory

Air Handling Unit (AHU) merupakan komponen utama dalam sistem pendingin udara (HVAC) kapal yang berfungsi untuk mengatur, mengondisikan, dan mendistribusikan udara ke seluruh ruangan. AHU bekerja dengan cara mencampur udara luar (fresh air) dan udara balik (return air), kemudian melewatkannya melalui filter, cooling coil, serta fan sebelum dialirkan ke ruang-ruang melalui ducting. Peran AHU sangat penting karena menentukan kenyamanan termal, kualitas udara dalam ruang, serta efisiensi energi sistem pendingin udara di kapal. Dalam perencanaan sistem, perhitungan kapasitas AHU harus didasarkan pada beban pendinginan total ruangan yang meliputi beban sensibel (perubahan suhu udara) dan beban laten (kelembapan udara).

Secara umum, kapasitas AHU dihitung menggunakan persamaan:

$$Q_{\text{total}} = \dot{m} (h_{\text{in}} - h_{\text{out}})$$

di mana Q_{total} adalah total kapasitas pendinginan (Watt), $\dot{m}$ adalah laju aliran massa udara (kg/s), sedangkan h_{in} dan h_{out} adalah entalpi udara masuk dan keluar koil pendingin (kJ/kg). Untuk menentukan laju aliran udara yang dibutuhkan AHU digunakan formula:

$$\dot{V} = \frac{Q_s}{\rho c_p \Delta T}$$

dengan $\dot{V}$ adalah laju aliran volumetrik udara (m³/s), ρ adalah densitas udara (1,2 kg/m³), c_p adalah kapasitas panas udara (1005 J/kg·K), dan ΔT adalah perbedaan suhu udara sebelum dan sesudah melewati koil. Selain itu, perhitungan daya fan pada AHU juga perlu dilakukan untuk memastikan udara dapat didorong melalui sistem ducting dengan tekanan yang cukup, menggunakan persamaan:

$$P_{\text{fan}} = \frac{\dot{V} \Delta p}{\eta_{\text{fan}}}$$

di mana P_{fan} adalah daya fan (Watt), Δp adalah total kehilangan tekanan pada sistem duct (Pa), dan η_{fan} adalah efisiensi fan. Dengan demikian, seluruh perencanaan AHU bertujuan agar udara yang dihasilkan memiliki temperatur dan kelembapan sesuai standar kenyamanan serta mampu menjangkau seluruh ruang kapal secara merata dan efisien.

2. REFERENCES

Sumber referensi dapat menggunakan sumber berikut :

1. Codes

Tuliskan code yang digunakan dalam desain (jika tidak ada dapat dihapus).

2. Standards

1. ASHRAE

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2. Handbook Of Air Conditioning System Design
3. Marine Engineering Handbook by R.L Harrington
4. SNAME4-7 2007
5. SNAME 4-16 2015

3. Technical documentation of the assets

1. General Arrangement M.T ANTARES 1
2. Hasil kalkulasi beban panas setiap ruangan M.T ANTARES 1

4. Other references

Tuliskan referensi-referensi lain seperti buku-buku atau jurnal paper di sini, dengan menggunakan **software Mendeley** (bisa download di [Download Mendeley Reference Manager For Desktop Windows | Mendeley](#) dan daftar dgn email @ppns) atau dengan software alternatif Zotero <https://www.zotero.org/download/>

Hasil seperti dibawah ini dan tampil secara otomatis :

- [1] J. M. C. Çengel, Yunus A, *Fluid Mechanics: Fundamentals and Applications*, 5th ed. New York: McGraw-Hill Education.
- [2] ASHRAE, *2018 Ashrae Handbook*. Atlanta, 2018. [Online]. Available: www.ashrae.org
- [3] "BITZER // Air Conditioning and Process Cooling." <https://www.bitzer.de/gb/en/air-conditioning-technology/> (accessed Dec. 07, 2022).
- [4] S. Klein and G. Nellis, *Mastering EES*. 2014.

3. DESIGN BASIS

3.1. Principal Dimension

Kapal yang akan dilakukan kalkulasi desain ini bernama M.T. Antares I. Kapal ini merupakan kapal niaga dengan jenis curah cair (oil product tanker) yang digunakan untuk mengangkut berbagai produk olahan minyak bumi atau bahan bakar cair. Kapal ini memiliki panjang keseluruhan sekitar 179,88 meter, lebar 32,20 meter, dengan daya angkut mati sebesar 45.997 ton, serta berbendera Liberia. Kapal ini dirancang untuk berlayar di perairan internasional, melayani distribusi produk minyak antar pelabuhan niaga. Adapun data utama kapal ini disajikan sebagai berikut. *Principal Dimension*

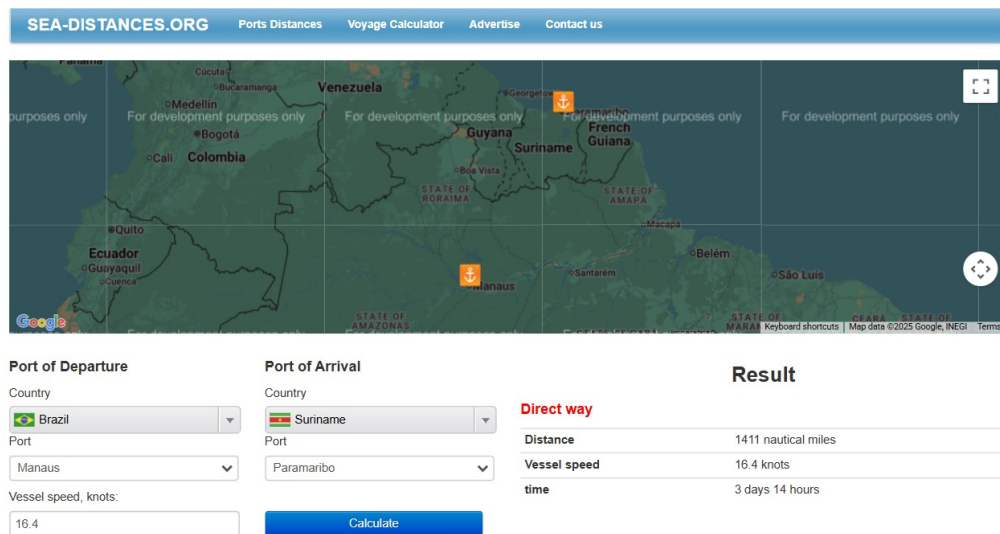
Tabel 3. 1 *Principal Dimension*

No.	Parameter	Nilai	Satuan
1.	<i>Length Perpendicular (LPP)</i>	172	Meter
2.	<i>Length Waterline (LWL)</i>	178.8	Meter
3.	<i>Breadth (B)</i>	32.2	Meter
4.	<i>Height (H)</i>	18.7	Meter
5.	<i>Draft (T)</i>	12.072	Meter
6.	<i>Coefficient Block (CB)</i>	0.726	
7.	<i>Coefficient Midship</i>	0.986	

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	(Cm)		
8.	Coefficient Prismatic (CP)	0.734	
9.	Kecepatan (vs)	16.4	Knot
10.	Radius Pelayaran	1411	Nautical Mills
11	Jumlah ABK	34	Orang

3.2. Sailing Route



Gambar 3. 1. Sailing Route

3.3. List of Room

Dalam mengondisikan udara pada ruang akomodasi di kapal, tentunya untuk mendapatkan perhitungan yang presisi perlu untuk mengetahui jumlah dan jenis ruangan yang udaranya akan dikondisikan. Pada perencanaan kali ini terdapat total 42 ruangan yang akan dikondisikan, dengan masing-masing 14 ruang di main deck, 9 ruang di poop deck, 8 ruang di boat deck, 6 ruang di bridge deck, dan ruang di navigation deck. Adapun tidak semua ruang pada kapal dilakukan pengkondisian udara. Karena juga terdapat ruang yang memang secara aturan tidak perlu atau bahkan dilarang untuk dilakukan pengkondisian udara. Maka dari itu, dibawah ini merupakan data daftar ruang-ruang yang akan dikondisikan pada tiap-tiap *deck* akomodasi yang ada pada kapal.

1. Main Deck

Tabel 3. 2 Daftar Ruangan yang dikondisikan di *Main Deck*

NO.	DECK	ROOM	P (m)	L (m)	T (m)	A (m2)	V (m3)
1	Main Deck	Mess Room	5.6	10.5	2.5	58.8	147

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2	Seaman (in Room 3 Person)	4.8	10.5	2.5	50.4	126
3	Fireman (in Room 3 Person)	4.8	10.5	2.5	50.4	126
4	Oilerman (in Room 3 Person)	4.8	10.5	2.5	50.4	126
5	Pumpman (in Room 4 Person)	4.8	10.5	2.5	50.4	126
6	Steward (in Room 2 person)	4.8	10.5	2.5	50.4	126
7	Assistant Cook (in Room 2 Person)	4.8	10.5	2.5	50.4	126
8	Boys (in Room 2 Person)	4	10.5	2.5	42	105
9	GYM Room	8	4.46	2.5	35.68	89.2
10	Entertainment Room	4	4.38	2.5	17.52	43.8
11	Clinic Room	5.4	4.38	2.5	23.652	59.13
12	Meeting Room	7.21	9	2.5	64.89	162.225
13	Workshop	3.8	4.4	2.5	16.72	41.8
14	Prayroom	3.2	4.4	2.5	14.08	35.2

2. Poop Deck

Tabel 3. 3 Daftar Ruangan yang dikondisikan di *Poop Deck*

1	Pray Room	4	6.5	2.5	26	65
2	Chief Cook	4	6.5	2.5	26	65
3	Boatswain	4.8	6.5	2.5	31.2	78
4	Electrician 1	4.8	6.5	2.5	31.2	78
5	Poop Deck Electrician 2	4.8	6.5	2.5	31.2	78
6	Electrician 3	4.8	6.5	2.5	31.2	78
7	Quarter Master	4	6.5	2.5	26	65
8	Hospital	4	3.6	2.5	14.4	36
9	GYM Room	4	3.6	2.5	14.4	36

3. Boat Deck

Tabel 3. 4 Daftar Ruangan yang dikondisikan di *Boat Deck*

1	Radio Operator 1	4.8	6.5	2.5	31.20	78.00
2	Radio Operator 2	4	6.5	2.5	26.00	65.00
3	Chief Officer (Mualim 1)	4.8	6.5	2.5	31.20	78.00
4	Boat Deck Second Officer (Mualim 2)	4.8	6.5	2.5	31.20	78.00
5	Second Engineer	4.8	6.5	2.5	31.20	78.00
6	Thrith Engineer	4.8	6.5	2.5	31.20	78.00
7	Doctor	4	6.5	2.5	26.00	65.00
8	Workshop	3.2	6.5	2.5	20.80	52.00

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4. Bridge Deck

Tabel 3. 5 Daftar Ruangan yang dikondisikan di *Bridge Deck*

	Bridge Deck						
1		<i>Pray Room</i>	4	6.5	2.5	26.00	65.00
2		<i>Captain Day Room</i>	3.2	6.5	2.5	20.80	52.00
3		<i>Captain Room</i>	6.4	6.5	2.5	41.60	104.00
4		<i>Chief Engineer</i>	6.4	6.5	2.5	41.60	104.00
5		<i>Chief Engineer Day Room</i>	3.2	6.5	2.5	20.80	52.00
6		<i>Entertainment Room</i>	4	6.5	2.5	26.00	65.00

5. Navigasi Deck

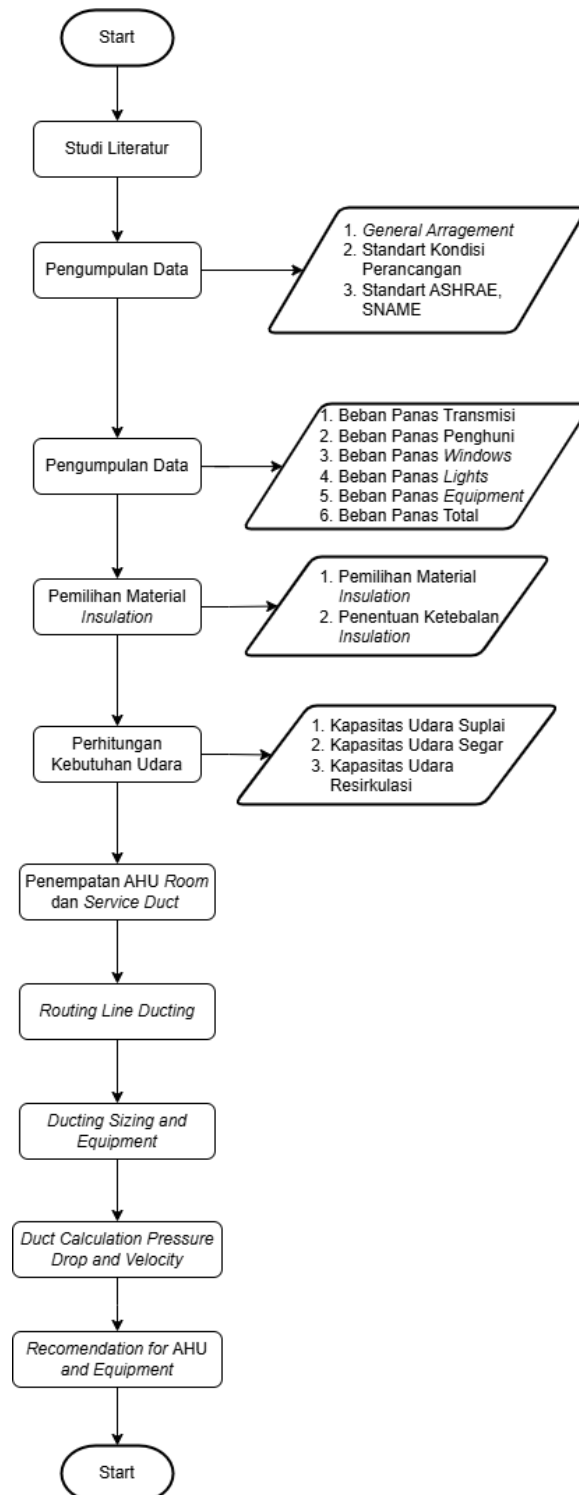
Tabel 3. 6 Daftar Ruangan yang dikondisikan di *Navigasi Deck*

	Navigation Deck						
1		<i>Pray Room</i>	1.6	5	2.5	8	20
2		<i>Chart Room</i>	2.4	5	2.5	12	30
3		<i>Radio Room</i>	2.4	5	2.5	12	30
4		<i>Meeting Room</i>	2.4	5	2.5	12	30
5		<i>Wheel House</i>	9.6	21	2.5	201.6	504

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4. METHODOLOGY

4.1. Diagram Alir Pengerjaan



Gambar 4. 1 Diagram Alir Pengerjaan

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4.2. Penjelasan Detail Flowchart

Flowchart tersebut menggambarkan metodologi pengerjaan proyek perencanaan sistem pendingin udara (Air Handling Unit dan ducting) pada kapal tanker, dengan alur kerja yang sistematis dari tahap awal hingga akhir.

1. Mulai
Merupakan titik awal dimulainya proyek perencanaan sistem pendingin udara kapal.
2. Studi Literatur & Data Tahap ini meliputi pengumpulan teori dasar tentang prinsip kerja AHU, sistem HVAC di kapal, serta standar yang digunakan seperti IMO, BKI, dan ASHRAE. Tujuannya agar perancangan sistem sesuai regulasi dan prinsip ilmiah.
3. Pengumpulan Data Kapal Menghimpun semua data yang diperlukan sebelum melakukan perhitungan teknis, meliputi :
 - a) GA (General Arrangement)
 - b) Data kondisi perancangan
 - c) Standar perancangan

4. Perhitungan Beban Panas

Menentukan total beban pendinginan (cooling load) yang harus ditangani oleh sistem tata udara. Perhitungannya mencakup:

- a) Beban panas transmisi
 - b) Beban Panas Penghuni
 - c) Beban Panas Jendela
 - d) Beban Panas Lampu
 - e) Beban Panas Peralatan
 - f) Beban Panas Total
5. Pemilihan Material Insulation dan Ketebalan Insulation yang sesuai dengan kebutuhan dapat mengurangi beban pendinginan/pemanasan, sehingga konsumsi daya listrik AHU semakin rendah
6. Perhitungan Kebutuhan Udara Menentukan kapasitas udara yang diperlukan untuk menjaga suhu dan kualitas udara ruangan. Perhitungannya meliputi:
 - a) Kapasitas Udara Suplai
 - b) Kapasitas Udara Segar
 - c) Kapasitas Udara Resirkulasi
7. Penentuan AHU Room, Service Duct dan spesifikasi AHU yang sesuai dengan hasil analisis kebutuhan. Tahap ini juga mencakup pemilihan cooling coil, fan, serta posisi pemasangan AHU di setiap deck agar efisien dan mudah diakses.
8. Duct Sizing and Equipment

Menentukan ukuran dan tipe ducting (saluran udara) serta memilih peralatan pendukung seperti:

- a) Diffuser, grille, damper, filter, dan fan.
 - b) Ukuran duct dihitung agar laju udara sesuai kecepatan standar dan tidak menimbulkan kebisingan.
9. Duct Calculation (Pressure Drop and Velocity)
Melakukan perhitungan pressure drop (kehilangan tekanan) dan kecepatan udara dalam duct:
 - a) Pressure drop dipengaruhi oleh panjang duct, belokan (elbow), fitting, dan permukaan saluran.
 - b) Hasil perhitungan digunakan untuk menentukan daya fan agar udara tetap mengalir sesuai kebutuhan.

10. Recommendation for AHU and Equipment

Tahap akhir berupa rekomendasi teknis yang mencakup:

- a) Kapasitas AHU (dalam CFM atau m^3/h).
 - b) Jenis fan, filter, dan komponen ducting.

11. Selesai

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Menandai bahwa seluruh proses perancangan sistem tata udara telah selesai dilakukan dan siap untuk tahap implementasi atau evaluasi.

5. DETAILED CALCULATION & RESULTS

Desain sistem pendingin mengacu pada SNAME 4-16 (2005) dimana ada beberapa beban panas yang ditimbulkan yang berpengaruh saat melakukan pengkondisian ruangan. Beberapa beban yang harus diperhitungkan adalah sebagai berikut:

Component	Section	Cooling Calculations	Heating Calculations	Ventilation Calculations
<i>Transmission</i>	3.1	X	X	X
<i>Solar</i>	3.2	X	-	X
<i>Lights</i>	3.3	X	X	X
<i>Equipment</i>	3.4	X	-	X
<i>Personnel</i>	3.5	X	-	-
<i>Infiltration</i>	3.7	-	X	-
<i>Ventilation</i>	3.9	-	-	X

Gambar 5. 1 Beban Panas Ruangan

5.1. Perhitungan Beban Panas dan Pemilihan Material Insulation beserta Ketebalan Insulation

5.1.1. Heat Load Transmition

Beban transmisi adalah aliran panas yang masuk melalui batas (sekat) karena perbedaan suhu di seluruh permukaan batas (sekat)

$$q = U \times A \times \Delta T$$

▲ T adalah perbedaan suhu di antara ruangan dan suhu ruangan sebelah. Setiap batas atau bagian yang memiliki nilai U atau ΔT yang berbeda harus dihitung secara terpisah. Jika suhu desain ruangan tidak tersedia, suhu yang tercatat harus digunakan atau perkiraan suhu ruangan bisa dihitung melalui pendekatan neraca panas yang disederhanakan. Nilai U harus merujuk pada versi terbaru dari Laporan Insulasi Termal, Buletin Teknis, dan Penelitian SNAME No. 4-7. Dalam perhitungan beban pendinginan, dampak pendinginan dari ruang tetangga diabaikan kecuali jika suhu yang lebih rendah dipertahankan oleh AC atau peralatan pendinginan. Sementara dalam perhitungan beban pemanasan, panas yang masuk melalui batas dikurangi hanya jika suhu yang lebih tinggi dari ruangan tetangga dijaga oleh sistem pemanas.

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Type of Space	T_i – Cooling (°F)	T_i – Heating (°F)
<i>Air Conditioning Equipment and Fan Rooms</i>	110	40
<i>Baggage</i>	110	60
<i>Battery Room</i>	110	40
<i>Bonded Stores</i>	110	40
<i>Bosn's Stores</i>	115	30
<i>Bow Thruster Room</i>	110	40
Type of Space	T_i – Cooling (°F)	T_i – Heating (°F)
<i>Butcher Shop</i>	105	40
<i>Canvas Room</i>	105	40
<i>Capstan Machinery Room</i>	110	30
<i>Cargo Control Room</i>	105	70
<i>Cargo Oil Tanks</i>	140	40
<i>Cargo Spaces Above Waterline</i>	120	20
<i>Cargo Spaces Below Waterline</i>	100	40
<i>Carpenter Shop</i>	105	60
<i>Cleaning Gear Locker</i>	105	60
<i>CO₂ Room</i>	110	40
<i>Deck Lockers, Deck Stores</i>	110	30
<i>Deck Toilet</i>	110	60
<i>Dry Stores</i>	110	50
<i>Elec. & Engrs. Workshops</i>	105	60
<i>Elevator Machinery Room</i>	110	40
<i>Emerg. Generator Room</i>	110	40
<i>Engrs. Stores</i>	115	30
<i>Fresh Water Tanks</i>	120	35
<i>F.O. Filling Station</i>	110	30
<i>F.O. Settling Tanks</i>	120	70
<i>F.O. Storage Tanks</i>	120	40

Gambar 5. 2 Temperatur Beban Panas

5.1.2. Solar and Transmision Load (Windows)

Beban transmisi surya + transmisi adalah aliran panas yang masuk akal melalui batas-batas cuaca ruangan yang terpapar matahari. Beban ini merupakan tambahan dari beban transmisi untuk batas atau batas-batas yang terpengaruh, tetapi beban total (surya + transmisi) dihitung dalam satu langkah.

$$q = U \times A \times \Delta T_e$$

Di mana ΔT adalah perbedaan antara suhu efektif dan suhu desain ruangan. Nilai U harus sesuai dengan standard SNAME 4-7 pada pemilihan jenis insulasi. Untuk beban solar yang melalui jendela dapat dihitung dengan persamaan:

$$q = G_{sf} \times A$$

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Jika lebih dari satu batas ruang terpapar sinar matahari, perhitungan perolehan panas yang terpisah harus dilakukan untuk setiap batas atau kombinasi batas dan perolehan simultan terbesar harus digunakan untuk menentukan beban panas matahari dan transmisi. Dimensi yang akan digunakan ketika menghitung area batas, kecuali bahwa area berbayang vertikal tidak boleh disertakan. Area yang diarsir harus dihitung dengan matahari pada sudut 45 dari matahari. Faktor beban matahari yang melalui jendela dan suhu efektif untuk perhitungan matahari. Faktor beban matahari yang melaluo jendela dan suhu efektif untuk perhitungan tunggal dan multi-batas tercantum dalam gambar 5.3 Jika suhu desain luar yang berbeda ditentukan, suhu pada gambar 5.3 harus disesuaikan, ke atas atau ke bawah.

<i>Glass Solar Factor</i>		Single Boundary Calculations		Multi-Boundary Calculations	
		160 Btu/hr/ft ²		120 Btu/hr/ft ²	
<i>Effective Temperatures</i>	Outdoor Air Dry Bulb Temperature	Single Boundary Calculations		Multi-Boundary Calculations	
		Horizontal Surface	Vertical Surface	Horizontal Surface	Vertical Surface
	90°F	140	120	125	110
	95°F	145	125	130	115
	100°F	150	130	135	120
	105°F	155	135	140	125
	110°F	160	140	145	130

Gambar 5. 3 Glass Solar Factor & Effective Temperatures

5.1.3. Equipment Load

Beban peralatan adalah panas sensibel dan laten yang dihasilkan oleh peralatan yang beroperasi di dalam ruang. Untuk ruangan ber-AC, komponen sensibel dan laten dari beban peralatan disertakan dalam. Namun, untuk ruang berventilasi, hanya komponen sensibel yang dimasukkan dalam perhitungan beban pendinginan. Ruang yang biasanya memiliki beban peralatan adalah dapur, pantry, binatu, ruang radio dan komunikasi, ruang kemudi, rumah resistor, kompartemen mesin geladak, dan ruang khusus seperti ruang komputer atau ruang kontrol mesin. Muatan peralatan biasanya akan disertakan untuk ruang tunggu, ruang mess.

$$q_s = q_{s,d} \times U.F. \times H.F.$$

Perolehan panas peralatan harus didasarkan pada data pembuangan panas untuk peralatan aktual yang dipasang. Sumber data pembuangan panas terbaik adalah pabrik pembuat peralatan; namun, sering kali peralatan belum dipilih saat perhitungan HVAC dilakukan, dalam hal ini sumber data pembuangan panas lainnya harus digunakan, seperti manual pabrik untuk peralatan serupa atau Buku Panduan Dasar ASHRAE. Gambar 5.5 disini memberikan data pembuangan panas untuk peralatan MARINE yang umum dan dapat digunakan ketika informasi yang lebih tepat tidak tersedia untuk desain tertentu.

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Equipment	Max. q_{sd}	Max. q_{ld}
<i>Clothes Dryer, Electric *</i>	4400	-
<i>Clothes Washer</i>	1300	-
<i>Coffee Maker</i>	6500	2000
<i>Coffee Warmer</i>	230	70
<i>Combination Steam Cooker/Kettle, Electric</i>	33500	17500
<i>Convection Oven, Electric</i>	13500	6900
<i>Dishwasher</i>	1120	580
<i>Fry Kettle, Electric</i>	12500	6900
<i>Garbage Disposal</i>	1120	-
<i>Griddle, Electric</i>	13500	6900
<i>Ice Cuber</i>	1300	-
<i>Iron, Electric</i>	3400	-

Table 3-5 Continued

Equipment	Max. q_{sd}	Max. q_{ld}
<i>Meat Slicer, Electric</i>	1120	580
<i>Mixer, Electric</i>	1120	580
<i>Oven, Electric</i>	13500	6900
<i>Range, Electric</i>	24700	12700
<i>Refrigerator (up to 20 cu.ft.)</i>	1300	-
<i>Toaster</i>	2230	1970

* Vented to Atmosphere

Gambar 5. 4 *Equipment Load*

5.1.4. Temperatur Desain Ruangan

Sistem pengkondisian udara pada ruang akomodasi di M.T Antares 1 yang meliputi main deck, poop deck, boat deck, bridge deck, dan navigation deck akan menggunakan temperatur acuan untuk mendesain seperti pada tabel berikut.

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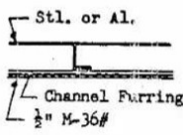
Tabel 5. 1 Desain Temperatur

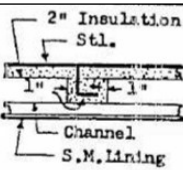
Desain	Temperture (°F)
Indoor (Inside Air) Dry Bulb	78
Outdoor (Weather Air) Dry Bulb	95
Indoor Gangway	80
Machinery Space	130
Vertical Steel Boundary	125
Horizontal Steel Boundary	145

5.1.5. Material Insulation Ruangan

Guna mendukung kinerja dari sistem pengkondisian udara pada ruang akomodasi di M.T Antares 1 yang meliputi main deck, poop deck, boat deck, bridge deck, dan navigation deck akan menggunakan insulasi untuk ruangan-ruangan tersebut dengan material insulasi dari SNAME. Adapun data heatflow dari SNAME dengan menggunakan material insulasi terdapat sebagai berikut.

Tabel 5. 2 Nilai Heat Flow SNAME sesuai Arah Perpindahan Panas

Type	Construstion	Condition		U	H	D
10		S	Solar Radiation	---	.495	.382
			Weather Air to Inside Air	.509	.425	.293
			Sea Water to Inside Air	---	.438	.292
			Inside Air to Inside Air	.435	.361	.272

Type	Construstion	Condition		U	H	D
55		S	Solar Radiation	---	.12	.11
			Weather Air to Inside Air	.11	.10	.10
			Sea Water to Inside Air	.11	.11	.10
			Inside Air to Inside Air	.11	.10	.10

5.2. Hasil Kalkulasi Beban Panas Tiap Ruangan `

5.2.1. Main Deck

Tabel 5. 3 Hasil Calculation Beban Panas Tiap Ruangan pada Main Deck

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SPACE	DECK	SIDE	FRAMES	Volume (ft ³)	Heat Area (ft ²)	BTU/HR			SHF
						SENS.	LATENT	TOTAL	
Mess Room	Main Deck	P	17 to 24	2088.16	1295.50	5718.29	2000	7718.285	0.741
Seaman (in Room 3 Person)		P	35 to 41	4449.65	2456.32	11111.32	1500	12611.32	0.88106
Fireman (in Room 3 Person)		P	29 to 35	4449.65	3129.82	11104.43	1500	12604.43	0.88099
Oiler (in Room 3 Person)		P	23 to 29	4449.65	3134.99	8904.24	1500	10404.24	0.856
Pumpman (in Room 4 Person)		P	17 to 23	4449.65	2390.93	8881.95	1500	10381.95	0.856
Steward (in Room 2 person)		S	35 to 41	4449.65	2454.26	9940.67	1000	10940.67	0.909
Assistant Cook (in Room 2 Person)		S	29 to 35	4449.65	2450.64	8429.16	1000	9429.163	0.894
Boys (in Room 2 Person)		S	24 to 29	4449.65	2450.64	8429.16	1000	9429.163	0.894
GYM Room		S	6 to 17	3150.07	1831.84	11078.53	3000	14078.53	0.787
Entertainment Room		S	-4 to 5	1546.78	1022.14	5535.08	2000	7535.083	0.735
Clinic Room		S	-4 to 5	2088.16	1295.50	5718.29	2000	7718.285	0.741
Meeting Room		C	(-5) to 10	5728.93	2274.73	19072.67	10000	29072.67	0.656
Workshop		P	5 to 11	1476.15	981.24	5930.97	2000	7930.967	0.748
Prayroom		P	11 to 15	1243.08	864.04	3243.53	2000	5243.533	0.619

5.2.2. Poop Deck

Tabel 5. 4 Hasil Calculate Beban Panas Tiap Ruangan pada Poop Deck

SPACE	DECK	SIDE	FRAMES	Volume (ft ³)	Heat Area (ft ²)	BTU/HR			SHF
						SENS.	LATENT	TOTAL	
Pray Room	Poop Deck	P	10 to 15	2295.46	1410.07	7230.83	2000	9230.827	0.783
Chief Cook		P	30 to 36	2295.46	876.72	4122.64	500	4622.64	0.89184
Boatswain		P	17 to 22	2754.55	1565.07	4116.01	500	4616.011	0.89168
Electrician 1		P	22 to 28	2754.55	1621.04	4040.29	500	4540.285	0.890
Electrician 2		S	30 to 36	2754.55	1621.04	4040.29	500	4540.285	0.890
Electrician 3		S	22 to 28	2754.55	1621.04	5366.06	1000	6366.055	0.843
Quarter Master		S	17 to 22	2295.46	1410.07	3972.63	1000	4972.626	0.799
Hospital		S	36 to 41	1271.33	879.41	3023.34	1000	4023.341	0.751
GYM Room		C	10 to 15	1271.33	876.72	8017.48	2000	10017.48	0.800

5.2.3. Boat Deck

Tabel 5. 5 Hasil Calculate Beban Panas Tiap Ruangan pada Boat Deck

SPACE	DECK	SIDE	FRAMES	Volume (ft ³)	Heat Area (ft ²)	BTU/HR			SHF
						SENS.	LATENT	TOTAL	
Radio Operator 1	Boat Deck	P	29 to 35	2754.55	1562.38	3132.11	500	3632.11	0.862
Radio Operator 2		P	24 to 29	2295.46	1407.38	2777.89	500	3277.892	0.84746
Chief Officer (Mualim 1)		S	35 to 41	2754.55	1618.35	4072.61	500	4572.612	0.89065
Second Officer (Mualim 2)		P	35 to 41	2754.55	1618.35	7262.19	500	7762.193	0.93559
Second Engineer		S	29 to 35	2754.55	1618.35	4072.61	500	4572.612	0.891
Thrith Engineer		S	23 to 29	2754.55	1618.35	3544.67	500	4044.67	0.876
Doctor		P	19 to 24	2295.46	1407.38	5748.08	500	6248.078	0.920
Workshop		S	19 to 23	1836.36	1196.41	5894.54	2000	7894.536	0.747

5.2.4. Bridge Deck

Tabel 5. 6 Hasil Calculate Beban Panas Tiap Ruangan pada Bridge Deck

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SPACE	DECK	SIDE	FRAMES	Volume (ft ³)	Heat Area (ft ²)	BTU/HR			SHF
						SENS.	LATENT	TOTAL	
Pray Room	Bridge Deck	P	24 to 29	2295.46	1407.38	6427.32	1000	7427.317	0.865
Captain Day Room		P	29 to 33	1836.36	1248.07	4053.05	500	4553.048	0.89018
Captain Room		P	33 to 41	2901.94	2008.95	8798.09	500	9298.085	0.946
Chief Engineer		S	33 to 41	3672.73	2329.85	11536.02	500	12036.02	0.958
Chief Engineer Day Room		S	29 to 33	1836.36	1196.41	5270.94	500	5770.94	0.913
Entertainment Room		S	24 to 29	2295.46	1235.16	9018.83	2000	11018.83	0.818

5.2.5. Navigasi Deck

Tabel 5. 7 Hasil Calculate Beban Panas Tiap Ruangan pada Navigasi Deck

SPACE	DECK	SIDE	FRAMES	Volume (ft ³)	Heat Area (ft ²)	BTU/HR			SHF
						SENS.	LATENT	TOTAL	
Pray Room	Navigation Deck	P	29 to 31	706.29	618.97	3253.06	1000	4253.057	0.765
Chart Room		P	34 to 37	1059.44	791.19	4191.47	1000	5191.468	0.807
Radio Room		S	34 to 37	1059.44	791.19	4213.43	1000	5213.433	0.808
Meeting Room		S	31 to 34	1059.44	1006.47	5113.99	2000	7113.995	0.719
Wheel House		C	29 to 41	12148.27	5058.02	30151.91	2000	32151.91	0.938

5.3. Hasil Calculate kebutuhan Suplai Udara Tiap Ruangan (Duct Airflow Calculation)

5.3.1. Main Deck

Tabel 5. 8 Hasil Calculate Kebutuhan Suplai Udara Tiap Ruangan pada Main Deck

SPACE	DECK	SIDE	FRAMES	Volume (ft ³)	Heat Area (ft ²)	OUTSIDE AIR BASED ON *			CFM
						R/C	PERS.	EXH.	
Mess Room	Main Deck	P	17 to 24	2088.16	1295.50	69.6052737	60	60	755
Seaman (in Room 3 Person)		P	35 to 41	4449.65	2456.32	148.32174	60	60	412
Fireman (in Room 3 Person)		P	29 to 35	4449.65	3129.82	148.32174	45	45	412
Oiler (in Room 3 Person)		P	23 to 29	4449.65	3134.99	148.32174	45	45	330
Pumpman (in Room 4 Person)		P	17 to 23	4449.65	2390.93	148.32174	72	72	329
Steward (in Room 2 person)		S	35 to 41	4449.65	2454.26	148.32174	30	30	369
Assistant Cook (in Room 2 Person)		S	29 to 35	4449.65	2450.64	148.32174	30	30	313
Boys (in Room 2 Person)		S	24 to 29	4449.65	2450.64	148.32174	30	30	270
GYM Room		S	6 to 17	3150.07	1831.84	105.0023747	90	90	411
Entertainment Room		S	-4 to 5	1546.78	1022.14	51.559462	60	60	206
Clinic Room		S	-4 to 5	2088.16	1295.50	69.6052737	60	60	212
Meeting Room		C	(-5) to 10	5728.93	2274.73	190.9642403	300	300	707
Workshop		P	5 to 11	1476.15	981.24	49.20514867	90	90	220
Prayroom		P	11 to 15	1243.08	864.04	41.43591467	60	60	186

5.3.2. Poop Deck

Tabel 5. 9 Hasil Calculate Kebutuhan Suplai Udara Tiap Ruangan pada Poop Deck

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SPACE	DECK	SIDE	FRAMES	Volume (ft ³)	Heat Area (ft ²)	OUTSIDE AIR BASED ON *			CFM
						R/C	PERS.	EXH.	
Pray Room	Poop Deck	P	10 to 15	2295.46	1410.07	76.51518333	60	60	268.00
Chief Cook		P	30 to 36	2295.46	876.72	76.51518333	15	15.00	153
Boatswain		P	17 to 22	2754.55	1565.07	91.81822	15	15	153
Electrician 1		P	22 to 28	2754.55	1621.04	91.81822	15	15	150.00
Electrician 2		S	30 to 36	2754.55	1621.04	91.81822	15	15	150.00
Electrician 3		S	22 to 28	2754.55	1621.04	91.81822	15	15	199.00
Quarter Master		S	17 to 22	2295.46	1410.07	76.51518333	15	15	148.00
Hospital		S	36 to 41	1271.33	879.41	42.37764	15	15	112.00
GYM Room		C	10 to 15	1271.33	876.72	42.37764	15	15	297.00

5.3.3. Boat Deck

Tabel 5. 10 Hasil Calculate Kebutuhan Suplai Udara Tiap Ruangan pada Boat Deck

SPACE	DECK	SIDE	FRAMES	Volume (ft ³)	Heat Area (ft ²)	OUTSIDE AIR BASED ON *			CFM
						R/C	PERS.	EXH.	
Radio Operator 1	Boat Deck	P	29 to 35	2754.55	1562.38	91.81822	15	15	117.00
Radio Operator 2		P	24 to 29	2295.46	1407.38	76.51518333	15	15	103
Chief Officer (Mualim 1)		S	35 to 41	2754.55	1618.35	91.81822	15	15	151
Second Officer (Mualim 2)		P	35 to 41	2754.55	1618.35	91.81822	15	15	269
Second Engineer		S	29 to 35	2754.55	1618.35	91.81822	15	15	151.00
Thrith Engineer		S	23 to 29	2754.55	1618.35	91.81822	15	15	132.00
Doctor		P	19 to 24	2295.46	1407.38	76.51518333	15	15	213.00
Workshop		S	19 to 23	1836.36	1196.41	61.21214667	60	60	219.00

5.3.4. Bridge Deck

Tabel 5. 11 Hasil Calculate Kebutuhan Suplai Udara Tiap Ruangan pada Bridge Deck

SPACE	DECK	SIDE	FRAMES	Volume (ft ³)	Heat Area (ft ²)	OUTSIDE AIR BASED ON *			CFM
						R/C	PERS.	EXH.	
Pray Room	Bridge Deck	P	24 to 29	2295.46	1407.38	76.51518333	30	30	239.00
Captain Day Room		P	29 to 33	1836.36	1248.07	61.21214667	15	15	151
Captain Room		P	33 to 41	2901.94	2008.95	96.73137764	15	15	326.00
Chief Engineer		S	33 to 41	3672.73	2329.85	122.4242933	15	15	428.00
Chief Engineer Day Room		S	29 to 33	1836.36	1196.41	61.21214667	15	15	196.00
Entertainment Room		S	24 to 29	2295.46	1235.16	76.51518333	15	15	335.00

5.3.5. Navigasi Deck

Tabel 5. 12 Hasil Calculate Kebutuhan Suplai Udara Tiap Ruangan pada Navigasi Deck

SPACE	DECK	SIDE	FRAMES	Volume (ft ³)	Heat Area (ft ²)	OUTSIDE AIR BASED ON *			CFM
						R/C	PERS.	EXH.	
Pray Room	Navigation Deck	P	29 to 31	706.29	618.97	23.54313333	30	30	121.00
Chart Room		P	34 to 37	1059.44	791.19	35.3147	30	30	156.00
Radio Room		S	34 to 37	1059.44	791.19	35.3147	30	30	157.00
Meeting Room		S	31 to 34	1059.44	1006.47	35.3147	60	60	190.00
Wheel House		C	29 to 41	12148.27	5058.02	404.9421876	75	75	1117.00

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5.4. Duct Material Selection

Duct Material Selection

Table 1 Duct Roughness Factors

Duct Material	Roughness Category	Absolute Roughness e , mm
Uncoated carbon steel, clean (Moody 1944) (0.05 mm)	Smooth	0.03
PVC plastic pipe (Swim 1982) (0.01 to 0.05 mm)		
Aluminum (Hutchinson 1953) (0.04 to 0.06 mm)		
Galvanized steel, longitudinal seams, 1200 mm joints (Griggs et al. 1987) (0.05 to 0.10 mm)	Medium smooth	0.09
Galvanized steel, continuously rolled, spiral seams, 3000 mm joints (Jones 1979) (0.06 to 0.12 mm)		
Galvanized steel, spiral seam with 1, 2, and 3 ribs, 3600 mm joints (Griggs et al. 1987) (0.09 to 0.12 mm)		
Galvanized steel, longitudinal seams, 760 mm joints (Wright 1945) (0.15 mm)	Average	0.15
Fibrous glass duct, rigid	Medium rough	0.9
Fibrous glass duct liner, air side with facing material (Swim 1978) (1.5 mm)		
Fibrous glass duct liner, air side spray coated (Swim 1978) (4.5 mm)	Rough	3.0
Flexible duct, metallic (1.2 to 2.1 mm when fully extended)		
Flexible duct, all types of fabric and wire (1.0 to 4.6 mm when fully extended)		
Concrete (Moody 1944) (1.3 to 3.0 mm)		

Class 1 - Stiffener/Weld Schedule
-2 to -30 in. wg

Spiral-Carbon & Galv

Negative Pressure, in wg		2		6		
Diameter, in.	Gauge	Stiffener Size/Type	No./Size of Welds	Gauge	Stiffener Size/Type	No./Size of Welds
4 thru 14	22	Unstiffened	22	22	Unstiffened	
16	20	Unstiffened	20	20	Unstiffened	
18	22	Unstiffened	18	18	Unstiffened	
20	22	Unstiffened	18	18	Unstiffened	
22	20	Unstiffened	18	18	Unstiffened	
24	20	Unstiffened	16	16	Unstiffened	
30	18	Unstiffened	20		L 1 x 1 x $\frac{1}{2}$	10 - 1.0
36	16	Unstiffened	20		L 1 $\frac{1}{2}$ x 1 $\frac{1}{2}$ x $\frac{1}{2}$	10 - 1.0
42	16	Unstiffened	20		L 1 $\frac{1}{2}$ x 1 $\frac{1}{2}$ x $\frac{1}{2}$	11 - 1.0
48	22	L 1 x 1 x $\frac{1}{2}$	11 - 1.0	20	L 1 $\frac{1}{2}$ x 1 $\frac{1}{2}$ x $\frac{1}{2}$	11 - 1.0
54	22	L 1 $\frac{1}{2}$ x 1 $\frac{1}{2}$ x $\frac{1}{2}$	12 - 1.0	20	L 2 x 2 x $\frac{1}{2}$	12 - 1.0
60	20	L 1 $\frac{1}{2}$ x 1 $\frac{1}{2}$ x $\frac{1}{2}$	12 - 1.0	18	L 2 x 2 x $\frac{1}{2}$	12 - 1.0
66	20	L 1 $\frac{1}{2}$ x 1 $\frac{1}{2}$ x $\frac{1}{2}$	13 - 1.0	18	L 2 x 2 x $\frac{1}{2}$	13 - 1.0
72	20	L 1 $\frac{1}{2}$ x 1 $\frac{1}{2}$ x $\frac{1}{2}$	13 - 1.0	18	L 2 x 2 x $\frac{1}{2}$	13 - 1.0
78	20	L 2 x 2 x $\frac{1}{2}$	13 - 1.0	16	L 2 x 2 x $\frac{1}{2}$	13 - 1.0
84	18	L 2 x 2 x $\frac{1}{2}$	14 - 1.0	16	L 2 x 2 x $\frac{1}{2}$	14 - 1.0
90	18	L 2 x 2 x $\frac{1}{2}$	14 - 1.0	16	L 2 x 2 x $\frac{1}{2}$	24 - 1.0
96	18	L 2 x 2 x $\frac{1}{2}$	14 - 1.0	16	L 2 x 2 x $\frac{1}{2}$	24 - 1.0

Berdasarkan standar Ashrae 2005, Chapter 35 Duct Design, Table 1 Duct Roughness Factors Untuk sistem distribusi HVAC sepanjang main deck sampai navigation deck dipilih **Uncoated carbon steel, clean (Moody 1944) (0.05 mm)** Pemilihan **uncoated carbon steel (clean)** didasarkan pada Duct Roughness Factors (Table 1) di mana material ini dikategorikan sebagai "smooth" dengan *absolute roughness* $e \approx 0.05$ mm. Nilai roughness ini menghasilkan friction factor dan loss pressure yang relatif rendah untuk duct round standar sehingga mendukung pemilihan diameter dan fan yang lebih ekonomis. Untuk menjaga kondisi "clean" sesuai asumsi tabel, akan diterapkan mitigasi korosi berupa field coating (primer anti-korosi + topcoat), pemasangan insulation/jacketing, drain/slope untuk condensate, dan inspeksi berkala, karena peningkatan roughness akibat korosi atau liner akan menaikkan pressure loss secara signifikan di

Materials

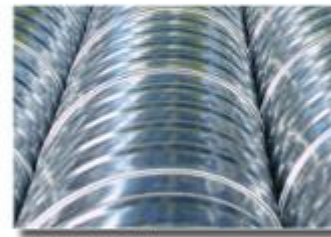
Galvanized steel, uncoated carbon steel, or aluminum are most frequently used for industrial air handling. Linx can also provide 304L and 316L stainless steel in a variety of finishes and aluminum steel for high temperature applications. Aluminum duct should not be used for conveying abrasive materials. When temperatures exceed 400°F (200°C) galvanized steel is not recommended. Please consult Linx for technical assistance in selecting material for handling corrosive gases, vapors, or mists.

Round Ducts

Duct and fittings are available in sizes ranging from 3" diameter to 96" diameter and material thickness from 26 gauge through 10 gauge.

Coordination Drawings

Linx has the in-house capability to generate detailed 3-D drawings to aid in the coordination of the fabrication and installation requirements. In addition, we can assist in monitoring and spooling sections of the ductwork at the factory to minimize field welding or bolted connections.



spiral lockseam duct

Gambar 2. 1 Pemilihan Duct Material

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5.5. DUCT SIZING AND SELECTION

5.5.1. DUCT SIZING BRANCH

- MAIN DECK (PORTSIDE LINE)

1	2	3	4	5	6	7	8	9
BRANCH SECT. NO.	AIR QUANTIT YQ (cfm)	EQUIV. DUCT LENGTH L (ft)	PRESSURE READING (in. wg)		TAKE-OFF TO TAKE OFF S.P. CHANGE (4 minus	S.P AHEAD OF TAKE OFF (in.wg)	AIR TERMINA L PRESS (in.wg)	DUCT SIZE (in.)
			Initial	Selected				
MD (A-01)	4920.00	33.6912			1.25	1.67114	1.5	14
			Duct Friction Loses		0.42114			
MD PS (01-02)	3007.00	48.8	1.1	1.12	-0.02	1.69114	1.5	12
MD PS (02-03)	2595.00	15.3832	0.86	0.86	0	1.69114	1.5	12
MD PS (03-04)	2183.00	15.6784	0.7	0.66	0.04	1.65114	1.5	12
MD PS (04-05)	1853.00	16.072	0.55	0.6	-0.05	1.70114	1.5	11
MD PS (05-06)	1524.00	18.368	0.46	0.39	0.07	1.63114	1.5	11
MD PS (06-07)	1338.00	5.412	0.35	0.32	0.03	1.60114	1.5	11
MD PS (07-08)	927.00	6.2648	0.27	0.23	0.04	1.56114	1.5	10
MD PS (08-09)	707.00	62.3472	0.085	0.07	0.015	1.54614	1.5	12
		Maximum S.P. at Section MD (06-07)				1.54614	+	1.5

2 diffuser Portside Line	CFM	Jumlah CFM	m ³ /h
206 Seaman Roor	412.00	3007.00	350.2
206 Fireman Roor	412.00	2595.00	350.2
165 Oilerman Roc	330.00	2183.00	280.5
164.5 Pumpman Ro	329.00	1853.00	279.65
Prayroom	186.00	1524.00	316.2
Gym Room	411.00	1338.00	698.7
Workshop	220.00	927.00	374
600.95 Meeting Roon	707.00		1201.9

Pipe Number	Uk. (in)	Uk. (m)
MD (A-01)	14	0.3556
MD PS (01-02)	12	0.3048
MD PS (02-03)	12	0.3048
MD PS (03-04)	12	0.3048
MD PS (04-05)	11	0.2794
MD PS (05-06)	11	0.2794
MD PS (06-07)	11	0.2794
MD PS (07-08)	10	0.254
MD PS (08-09)	12	0.3048

Urutan Section	cfm
MD (A-01)	4920.00
MD PS (01-02)	3007.00
MD PS (02-03)	2595.00
MD PS (03-04)	2183.00
MD PS (04-05)	1853.00
MD PS (05-06)	1524.00
MD PS (06-07)	1338.00
MD PS (07-08)	927.00
MD PS (08-09)	707.00

Pipe Length	m	ft
MD (A-01)	6	19.68
MD PS (01-02)	10	32.8
MD PS (02-03)	4.69	15.3832
MD PS (03-04)	4.78	15.6784
MD PS (04-05)	4.9	16.072
MD PS (05-06)	5.6	18.368
MD PS (06-07)	1.65	5.412
MD PS (07-08)	1.91	6.2648
MD PS (08-09)	14.74	48.3472

Gambar 2. 2 Duct Sizing Branch Maindeck (Portside Line)

- MAIN DECK (STARBOARD LINE)

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1	2	3	4	5	6	7	8	9
BRANCH SECT. NO.	AIR QUANTIT Y Q (cfm)	EQUIV. DUCT LENGTH L (ft)	PRESSURE READING (in. wg)		TAKE-OFF TO TAKE OFF S.P. CHANGE (4 minus	S.P AHEAD OF TAKE OFF (in.wg)	AIR TERMINA L PRESS (ib.wg)	DUCT SIZE (in.)
			Initial	Selected				
MD (A-01)	4920	33.6912			1.25	1.67114	1.5	14
			Duct Friction Loses		0.42114			
MD SB (01-10)	1913.00	44.8	1.15	0.92	0.23	1.44114	1.5	10
MD SB (10-11)	1544.00	15.6128	0.75	0.62	0.13	1.31114	1.5	10
MD SB (11-12)	1231.00	14.76	0.5	0.4	0.1	1.21114	1.5	10
MD SB (12-13)	961.00	15.58	0.33	0.4	-0.07	1.28114	1.5	9
MD SB (13-14)	206.00	20.992	0.24	0.24	0	1.28114	1.5	5
Maximum S.P. at Section MD (06-07)						1.28114	+	1.5

Starboard Line	CFM	Jumlah	2 Diffuser m ³ /h
Steaward	369.00	1913.00	184.5
Assistant Cook	313.00	1544.00	156.5
Boys	270.00	1231.00	135
Meshroom	755.00	961.00	377.5
Entertainment Room	206.00		350.2

Urutan Section	cfm
MD SB (01-10)	1913.00
MD SB (10-11)	1544.00
MD SB (11-12)	1231.00
MD SB (12-13)	961.00
MD SB (13-14)	206.00

Dimension Branch Round Duct

Pipe Number	Uk. (in)	Uk. (m)	Pipe Length	m	ft
MD SB (01-10)	10	0.254	MD SB (01-10)	10	32.8
MD SB (10-11)	10	0.254	MD SB (10-11)	4.76	15.6128
MD SB (11-12)	10	0.254	MD SB (11-12)	4.5	14.76
MD SB (12-13)	9	0.2286	MD SB (12-13)	4.75	15.58
MD SB (13-14)	5	0.127	MD SB (13-14)	6.4	20.992

Gambar 2. 3 Duct Sizing Branch Maindeck (Starboard Line)

- POOP DECK (PORTSIDE LINE)

AIR HANDLING UNIT DETAILED DESIGN Calculation Reports & Drawings for MT. ANTARES 1			 PPNS POLITEKNIK PONTIANAK NEGERI SEMBILAN	Prep. :	HMP
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1	2	3	4	5	6	7	8	9	10	11	12
BRANCH SECT. NO.	AIR QUANTIT Y Q (cfm)	EQUIV. DUCT LENGTH L (ft)	PRESSURE READING (in. wg)		TAKE-OFF TO TAKE OFF S.P. CHANGE (4 minus)	S.P AHEAD OF TAKE OFF (in.wg)	AIR TERMINA L PRESS (in.wg)	DUCT SIZE (in.)	VELOCITY V (fpm)	Loses Branch to Terminal (in.wg)	DUCT Terminal Size (in.)
			Initial	Selected							
PD PS(B-01)	724.00	56.164			1.25	3.384232	1.5	5	4500	0.16	5
			Duct Friction Loss =		2.134232						
PD PS(01-02)	571.00	20.664	0.68	0.88	-0.2	3.584232	1.5	6	3100	0.16	5
PD PS(02-03)	421.00	14.76	0.52	0.45	0.07	3.514232	1.5	6	2200	0.16	5
PD PS(03-04)	268.00	18.696	0.32	0.42	-0.1	3.614232	1.5	5	2000	0.45	5
		Maximum S.P. at Section MD (06-07)				3.384232	+	1.5	+	0.16	=

Portside Line	CFM	Jumlah CFM	m ³ /h
Chief Cook	153.00	724.00	260.1
Electrician 1	150.00	571.00	255
Boatswain	153.00	421.00	260.1
Mushola	268.00		455.6

Urutan Section	cfm
PD PS(B-01)	724.00
PD PS(01-02)	571.00
PD PS(02-03)	421.00
PD PS(03-04)	268.00

Dimension Round Duct

Pipe Number	Uk. (in)	Uk. (m)
PD PS(B-01)	5	0.127
PD PS(01-02)	6	0.1524
PD PS(02-03)	6	0.1524
PD PS(03-04)	5	0.127

Pipe Length	m	ft
PD PS(B-01)	9.5	31.16
PD PS(01-02)	6.3	20.664
PD PS(02-03)	4.5	14.76
PD PS(03-04)	5.7	18.696

Gambar 2. 4 Duct Sizing Branch Poopdeck (Portside Line)

- POOP DECK (STARBOARD LINE)

1	2	3	4	5	6	7	8	9	10	11	12
BRANCH SECT. NO.	AIR QUANTIT Y Q (cfm)	EQUIV. DUCT LENGTH L (ft)	PRESSURE READING (in. wg)		TAKE-OFF TO TAKE OFF S.P. CHANGE (4 minus)	S.P AHEAD OF TAKE OFF (in.wg)	AIR TERMINA L PRESS (ib.wg)	DUCT SIZE (in.)	VELOCITY V (fpm)	Loses Branch to Terminal (in.wg)	DUCT Terminal Size (in.)
			Initial	Selected							
PD SB(B-05)	497.00	42.7632			1.25	3.38816	1.5	4	4500	0.16	5
			Duct Friction Loss =		2.13816						
PD SB(05-06)	347.00	20.664	0.42	0.47	-0.05	3.43816	1.5	5	1800	0.28	5
PD SB(06-07)	148.00	14.76	0.19	0.14	0.05	3.38816	1.5	5	1100	0.16	5
		Maximum S.P. at Section MD (06-07)				3.43816	+	1.5	+	0.28	=

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Starboard Line	CFM	Jumlah	m ³ /h
Electrician 2	150.00	497.00	255
Electrician 3	199.00	347.00	338.3
Quarter Master	148.00	148.00	251.6

Urutan Section	cfm
PD SB (B-05)	497.00
PD SB (05-06)	347.00
PD SB (06-07)	148.00

Dimension Round Duct		
Pipe Number	Uk. (in)	Uk. (m)
PD SB (B-05)	4	0.1016
PD SB (05-06)	5	0.127
PD SB (06-07)	5	0.127

Pipe Length	m	ft
PD SB (B-05)	9.5	31.16
PD SB (05-06)	6.3	20.664
PD SB (06-07)	4.5	14.76

Gambar 2. 5 Duct Sizing Branch Poopdeck (Starboard Line)

- BOAT DECK (PORTSIDE LINE)

1	2	3	4	5	6	7	8	9	10	11	12
BRANCH SECT. NO.	AIR QUANTIT Y Q (cfm)	EQUIV. DUCT LENGTH L (ft)	PRESSURE READING (in. wg)		TAKE-OFF TO TAKE OFF S.P. CHANGE (4 minus	S.P AHEAD OF TAKE OFF (in.wg)	AIR TERMINA L PRESS (in.wg)	DUCT SIZE (in.)	VELOCITY V (fpm)	Loses Branch to Terminal (in.wg)	DUCT Terminal Size (in.)
			Initial	Selected							
BD C (C-01)	1355.00	28.2272			1.25	2.040362	1.5	7	4500		
			Duct Friction Loss =			0.790362					
BD PS (01-02)	702.00	30.6	0.9	0.68	0.22	1.820362	1.5	7	2700	0.225	6
BD PS (02-03)	433.00	15.42	0.45	0.45	0	1.820362	1.5	6	2200	0.275	4
BD PS (03-04)	316.00	14.76	0.29	0.27	0.02	1.800362	1.5	6	1700	0.19	4
BD PS (04-05)	213.00	12.464	0.17	0.125	0.045	1.755362	1.5	6	1100	0.3	5
		Maximum S.P. at Section MD (06-07)				1.820362	=	1.5	+	0.225	=

Portside Line	CFM	Jumlah CFM	m ³ /h
Second Office	269.00	702.00	457.3
Radio Operator	117.00	433.00	198.9
Radio Operator	103.00	316.00	175.1
Doctor	213.00		362.1

Urutan Section	cfm
BD C (C-01)	1355.00
BD PS (01-02)	702.00
BD PS (02-03)	433.00
BD PS (03-04)	316.00
BD PS (04-05)	213.00

Dimension Round Duct		
Pipe Number	Uk. (in)	Uk. (m)
BD C (C-01)	7	0.1778
BD PS (01-02)	7	0.1778
BD PS (02-03)	6	0.1524
BD PS (03-04)	6	0.1524
BD PS (04-05)	6	0.1524

Pipe Number	m	ft
BD C (C-01)	6.47	21.2216
BD PS (01-02)	7.5	24.6
BD PS (02-03)	4.7	15.416
BD PS (03-04)	4.5	14.76
BD PS (04-05)	3.8	12.464

Gambar 2. 6 Gambar 2. 7 Duct Sizing Branch Boatdeck (Portside Line)

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- BOAT DECK (STARBOARD LINE)

1	2	3	4	5	6	7	8	9	10	11	12
BRANCH SECT. NO.	AIR QUANTIT Y Q (cfm)	EQUIV. DUCT LENGTH L (ft)	PRESSURE READING (in. wg)		TAKE-OFF TO TAKE OFF S.P. CHANGE (4 minus)	S.P AHEAD OF TAKE OFF (in.wg)	AIR TERMINA L PRESS (ib.wg)	DUCT SIZE (in.)	VELOCITY V (fpm)	Loses Branch to Terminal (in.wg)	DUCT Terminal Size (in.)
			Initial	Selected							
BD C (C-01)	1355	28.2272	0	0	1.25	2.040362	1.5	7	4500		
			Duct Friction Loss =			0.790362					
BD SB (01-06)	653.00	30.6	0.95	0.76	0.19	1.850362	1.5	7	2500	0.16	5
BD SB (06-07)	502.00	15.416	0.36	0.34	0.02	1.830362	1.5	7	1900	0.16	5
BD SB (07-08)	351.00	15.744	0.22	0.28	-0.06	1.890362	1.5	6	1700	0.12	5
BD SB (08-09)	219.00	13.12	0.17	0.12	0.05	1.840362	1.5	6	1100	0.16	6
		Maximum S.P. at Section MD (06-07)				1.830362	+	1.5	+	0.16	=

Starboard Line	CFM	Jumlah	m ³ /h
Chief Officer 1 (Mualim	151.00	653.00	256.7
Second Engineer	151.00	502.00	256.7
Thrith Engineer	132.00	351.00	224.4
Workshop	219.00		372.3

Urutan Section	cfm
BD SB (01-06)	653.00
BD SB (06-07)	502.00
BD SB (07-08)	351.00
BD SB (08-09)	219.00

Dimension Round Duct		
Pipe Number	Uk. (in)	Uk. (m)
BD SB (01-06)	7	0.1778
BD SB (06-07)	7	0.1778
BD SB (07-08)	6	0.1524
BD SB (08-09)	6	0.1524

Pipe Number	m	ft
BD SB (01-06)	7.5	24.6
BD SB (06-07)	4.7	15.416
BD SB (07-08)	4.8	15.744
BD SB (08-09)	4	13.12

Gambar 2. 8 Duct Sizing Branch Boatdeck (Starboard Line)

- BRIDGE DECK (PORTSIDE LINE)

1	2	3	4	5	6	7	8	9	10	11	12
BRANCH SECT. NO.	AIR QUANTIT Y Q (cfm)	EQUIV. DUCT LENGTH L (ft)	PRESSURE READING (in. wg)		TAKE-OFF TO TAKE OFF S.P. CHANGE (4 minus)	S.P AHEAD OF TAKE OFF (in.wg)	AIR TERMINA L PRESS (in.wg)	DUCT SIZE (in.)	VELOCITY V (fpm)	Loses Branch to Terminal (in.wg)	DUCT Terminal Size (in.)
			Initial	Selected							
BGD C (D-01)	1675.00	31.2296			1.25	2.03074	1.5	10	4500		
			Duct Friction Loss =			0.78074					
BGD PS (01-02)	716.00	35.24	0.8	0.68	0.12	1.91074	1.5	7	2700	0.475	8
BGD PS (02-03)	390.00	16.4	0.41	0.41	0	1.91074	1.5	6	2000	0.55	6
BGD PS (03-04)	239.00	11.808	0.27	0.33	-0.06	1.97074	1.5	5	1800	0.35	5
		Maximum S.P. at Section BGD SB (03-04)				1.97074	+	1.5	+	0.35	=

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Portside Line	CFM	Jumlah CFM	m ³ /h
Captain Room	326.00	716.00	554.2
Captain Day Room	151.00	390.00	256.7
Prayroom	239.00		406.3

Urutan Section	cfm
BGD C (D-01)	1675.00
BGD PS (01-02)	716.00
BGD PS (02-03)	390.00
BGD PS (03-04)	239.00

Dimension Round Duct		
Pipe Number	Uk. (in)	Uk. (m)
BGD C (D-01)	10	0.254
BGD PS (01-02)	7	0.1778
BGD PS (02-03)	6	0.1524
BGD PS (03-04)	5	0.127

Pipe Number	m	ft
BGD C (D-01)	6.47	21.2216
BGD PS (01-02)	8	26.24
BGD PS (02-03)	5	16.4
BGD PS (03-04)	3.6	11.808

Gambar 2. 9 Duct Sizing Branch Bridge Deck (Portside Line)

- BRIDGE DECK (STARBOARD LINE)

1	2	3	4	5	6	7	8	9	10	11	12
BRANCH SECT. NO.	AIR QUANTIT Y Q (cfm)	EQUIV. DUCT LENGTH L (ft)	PRESSURE READING (in. wg)		TAKE-OFF TO TAKE OFF S.P. CHANGE (4 minus)	S.P AHEAD OF TAKE OFF (in.wg)	AIR TERMINA L PRESS (ib.wg)	DUCT SIZE (in.)	VELOCITY V (fpm)	Losses Branch to Terminal (in.wg)	DUCT Terminal Size (in.)
			Initial	Selected							
BGD C (D-01)	2581	31.2296			1.25	2.03074	1.5	10	4500		
			Duct Friction Loss =		0.78074						
BGD SB (01-0)	959.00	35.24	0.1	0.15	-0.05	2.08074	1.5	7	3500	0.3	7
BGD SB (02-0)	531.00	16.4	1.1	1.15	-0.05	2.03074	1.5	7	2100	0.275	5
BGD SB (03-0)	335.00	11.808	0.26	0.3	-0.04	2.07074	1.5	6	1700	0.375	6
		Maximum S.P. at Section BGD SB (03-04)				2.07074	+	1.5	+	0.375	=

Starboard Line	CFM	Jumlah	m ³ /h
Chief Engineer Room	428.00	959.00	727.6
Chief Engineer Day Room	196.00	531.00	333.2
Entertainment Room	335.00		569.5

Urutan Section	cfm
BGD SB (01-05)	959.00
BGD SB (05-06)	531.00
BGD SB (06-07)	335.00

Dimension Round Duct		
Pipe Number	Uk. (in)	Uk. (m)
BGD SB (01-05)	7	0.1778
BGD SB (05-06)	5	0.127
BGD SB (06-07)	6	0.1524

Pipe Number	m	ft
BGD C (D-01)	6.47	21.2216
BGD SB (01-0)	8	26.24
BGD SB (05-0)	5	16.4
BGD SB (06-0)	3.6	11.808

Gambar 2. 10 Duct Sizing Branch Bridge Deck (Starboard Line)

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- NAVIGASI DECK (PORTSIDE LINE)

1	2	3	4	5	6	7	8	9	10	11	12
BRANCH SECT. NO.	AIR QUANTIT Y Q (cfm)	EQUIV. DUCT LENGTH L (ft)	PRESSURE READING (in. wg)		TAKE-OFF TO TAKE OFF S.P. CHANGE (4 minus)	S.P AHEAD OF TAKE OFF (in.wg)	AIR TERMINA L PRESS (in.wg)	DUCT SIZE (in.)	VELOCITY V (fpm)	Loses Branch to Terminal (in.wg)	DUCT Terminal Size (in.)
			Initial	Selected							
ND C (E-01)	1741.00	21.848			1.25	1.44348	1.5	8	4500		
			Duct Friction Loss =		0.19348						
ND PS (01-02)	835.50	11.48	1.35	1.7	-0.35	1.79348	1.5	6	4300	0.5	7
ND PS (02-03)	277.00	15.28	1.25	1.05	0.2	1.99348	1.5	4	3200	0.425	4
ND PS (03-04)	121.00	14.432	0.62	0.22	0.4	2.39348	1.5	4	1300	0.275	4
Maximum S.P. at Section BGD SB (03-04)						1.79348	+	1.5	+	0.5	=

Portside Line	CFM	Jumlah CFM	m ³ /h
WheelHouse	558.5	835.50	949.45
Chaft Room	156.00	277.00	265.2
Mushola	121.00		205.7

Urutan Section	cfm
ND C (E-01)	1741.00
ND PS (01-02)	835.50
ND PS (02-03)	277.00
ND PS (03-04)	121.00

Dimension Round Duct		
Pipe Number	Uk. (Inchi)	Uk. (m)
ND C (E-01)	8	0.2032
ND PS (01-02)	6	0.1524
ND PS (02-03)	4	0.1016
ND PS (03-04)	4	0.1016

Pipe Number	m	ft
ND C (E-01)	4.22	13.8416
ND PS (01-02)	3.5	11.48
ND PS (02-03)	3.5	11.48
ND PS (03-04)	4.4	14.432

Gambar 2. 11 Duct Sizing Branch Navigasi Deck (Portside Line)

- NAVIGASI DECK (STARBOARD LINE)

1	2	3	4	5	6	7	8	9	10	11	12
BRANCH SECT. NO.	AIR QUANTIT Y Q (cfm)	EQUIV. DUCT LENGTH L (ft)	PRESSURE READING (in. wg)		TAKE-OFF TO TAKE OFF S.P. CHANGE (4 minus)	S.P AHEAD OF TAKE OFF (in.wg)	AIR TERMINA L PRESS (ib.wg)	DUCT SIZE (in.)	VELOCITY V (fpm)	Loses Branch to Terminal (in.wg)	DUCT Terminal Size (in.)
			Initial	Selected							
ND C (E-01)	1741	21.848	0	0	1.25	1.44348	1.5	8	4500		0
			Duct Friction Loss =		0.19348						
ND SB (01-05)	905.50	11.48	1.35	1.05	0.3	1.14348	1.5	7	3400	0.5	7
ND SB (05-06)	347.00	15.98	0.7	0.7	0	1.14348	1.5	5	2600	0.425	4
ND SB (06-07)	190.00	7.216	0.4	0.59	-0.19	0.95348	1.5	4	2300	0.7	4
Maximum S.P. at Section BGD SB (03-04)						1.14348	+	1.5	+	0.5	=

Starboard Line	CFM	Jumlah	m ³ /h
WheelHouse	558.50	905.50	949.45
Radio Room	157.00	347.00	266.9
Meeting Room	190.00	190.00	323

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Urutan Section	cfm
ND SB (01-05)	905.50
ND SB (05-06)	347.00
ND SB (06-07)	190.00

Dimension Round Duct		
Pipe Number	Uk. (Inchi)	Uk. (m)
ND C (E-01)	8	0.2032
ND SB (01-05)	7	0.1778
ND SB (05-06)	5	0.127
ND SB (06-07)	4	0.1016

Pipe Length	m	ft
ND SB (01-05)	3.5	11.48
ND SB (05-06)	3.5	11.48
ND SB (06-07)	2.2	7.216

Gambar 2. 12 Duct Sizing Branch Navigasi Deck (Starboard Line)

5.5.2. Duct Sizing Header (Lower Deck)

1	2	3	4	5	6	7	8	9	10
HEADER SECT. NO.	AIR QUANTITY Q (cfm)	EQUIV. DUCT LENGTH L (ft)	PRESSURE READING (in. wg)		TAKE-OFF TO TAKE OFF S.P. CHANGE (4 minus	S.P AHEAD OF TAKE OFF (in.wg)	BRANCH PRESS. (in.wg)	DUCT SIZE (in.)	VELOCITY V (fpm)
			Initial	Selected					
M. Header Ex.	10415.00	28.8072				0.282		22	3500
MD (A)	4920.00	3.7392	0.9	0.78	0.12	0.162	3.37114	18	3100

Portside Line	CFM	Jumlah CFM
Main Deck	4920.00	4920.00

Urutan Section	cfm
M. Header Ex. AHU	10415.00
MD (A)	4920.00

Dimension Round Duct		
Pipe Number	Uk. (Inchi)	Uk. (m)
M. Header Ex. AHU	22	0.5588
MD (A)	18	0.4572

Pipe Number	m	ft
M. Header Ex.	2.07	6.7896
MD (A)	1.14	3.7392

Gambar 2. 13 Duct Sizing Header (Lower Deck Line)

5.5.3. Duct Sizing Header (Upper Deck)

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1	2	3	4	5	6	7	8	9	10
BRANCH SECT. NO.	AIR QUANTIT Y Q (cfm)	EQUIV. DUCT LENGTH L (ft)	PRESSURE READING (in. wg)		TAKE-OFF TO TAKE OFF S.P. CHANGE (4 minus)	S.P AHEAD OF TAKE OFF (in.wg)	BRANCH PRESS. (in.wg)	DUCT SIZE (in.)	VELOCITY V (fpm)
			Initial	Selected					
M. Header Ex	10415	28.8072				0.282072		22	3500
PD (B)	5495.00	4.1984	0.9	0.78	18	3100	5.564232	18	3100
BD (C)	4771.00	8.2	0.59	0.525	0.065	3099.935	3.595362	18	2600
BGD (D)	3416.00	8.2	0.425	0.5	-0.075	3100.01	3.96074	16	2400
ND (E)	1741.00	8.2	0.4	0.45	-0.05	3100.06	4.16848	12	2100
Maximum S.P. at Section BGD SB (03-04)						3100	+	5.564232	+

Starboard Line	CFM	Jumlah
Poop Deck	724.00	5495.00
Boat Deck	1355.00	4771.00
Bridge Deck	1675.00	3416.00
Navigasi Deck	1741.00	

Urutan Section	cfm
PD (B)	5495.00
BD (C)	4771.00
BGD (D)	3416.00
ND (E)	1741.00

Dimension Round Duct			
Pipe Number	Uk. (Inchi)	Uk. (m)	
PD (B)	18	0.4572	
BD (C)	18	0.4572	
BGD (D)	16	0.4064	
ND (E)	12	0.3048	

Pipe Length	m	ft
PD (B)	1.28	4.1984
BD (C)	2.5	8.2
BGD (D)	2.5	8.2
ND (E)	2.5	8.2

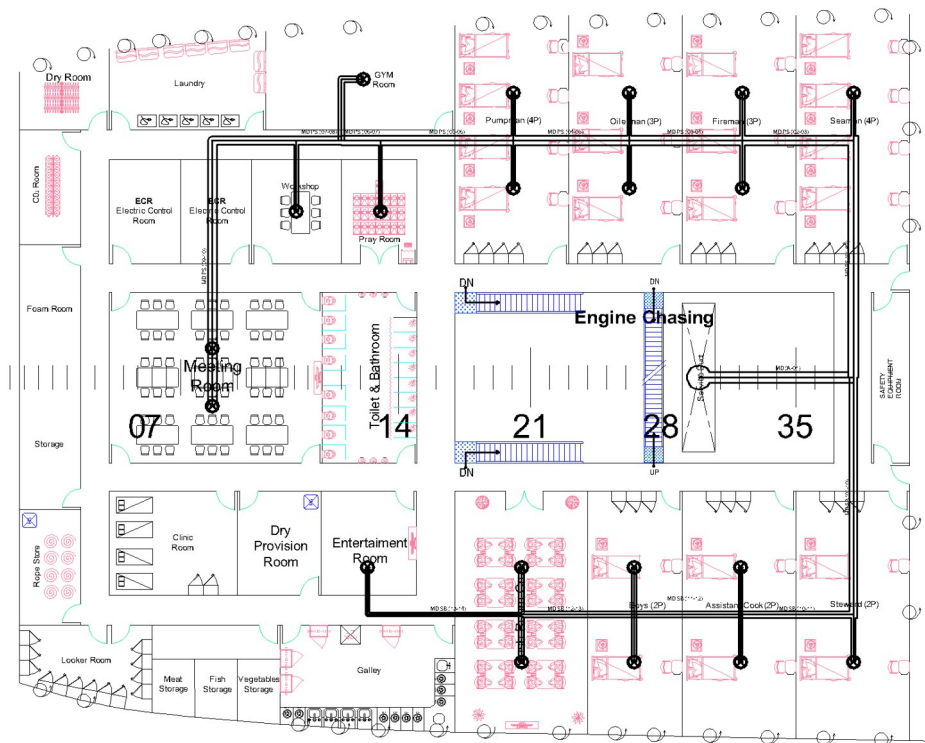
Gambar 2. 14 Duct Sizing Header (Upper Deck Line)

5.6. INSULATION MATERIAL SELECTION AND THICKNESS DETERMINATION

5.7. DUCT ROUTING

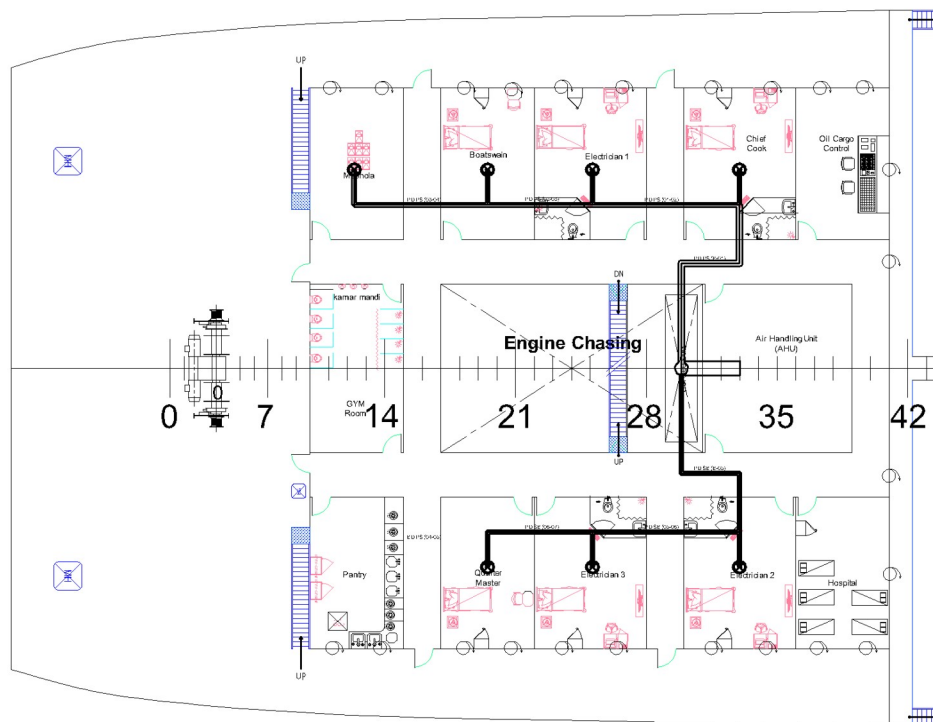
- Main Deck

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Gambar 5. 5 Duct Routing Maindeck

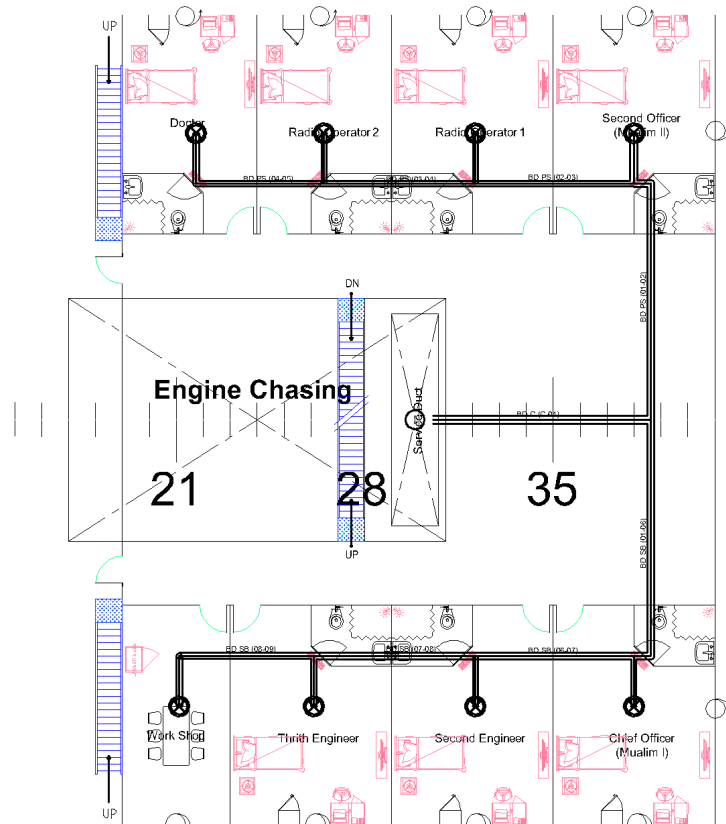
- Poop Deck



Gambar 5. 6 Duct Routing Poop Deck

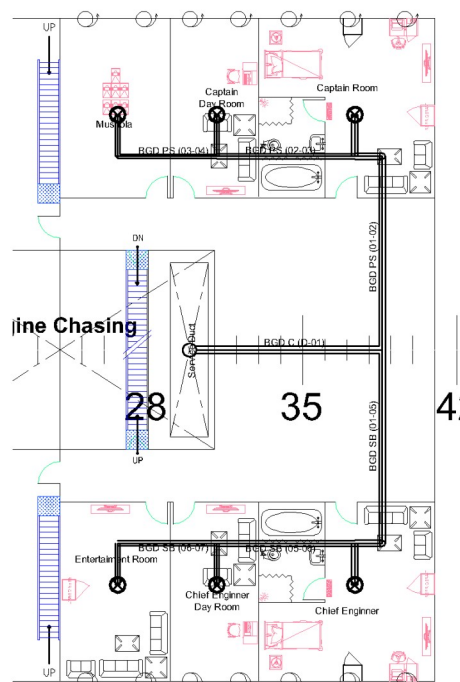
- Boat Deck

AIR HANDLING UNIT DETAILED DESIGN Calculation Reports & Drawings for MT. ANTARES 1			Prep. :	HMP
			Chk. :	GEK
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Gambar 5. 7 Duct Routing Boat Deck

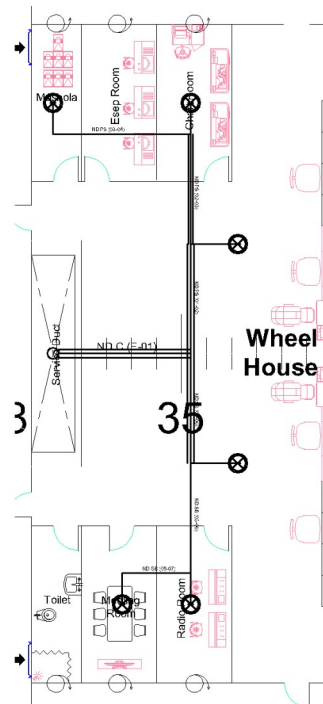
- Bridge Deck



Gambar 5. 8 Duct Routing Bridge Deck

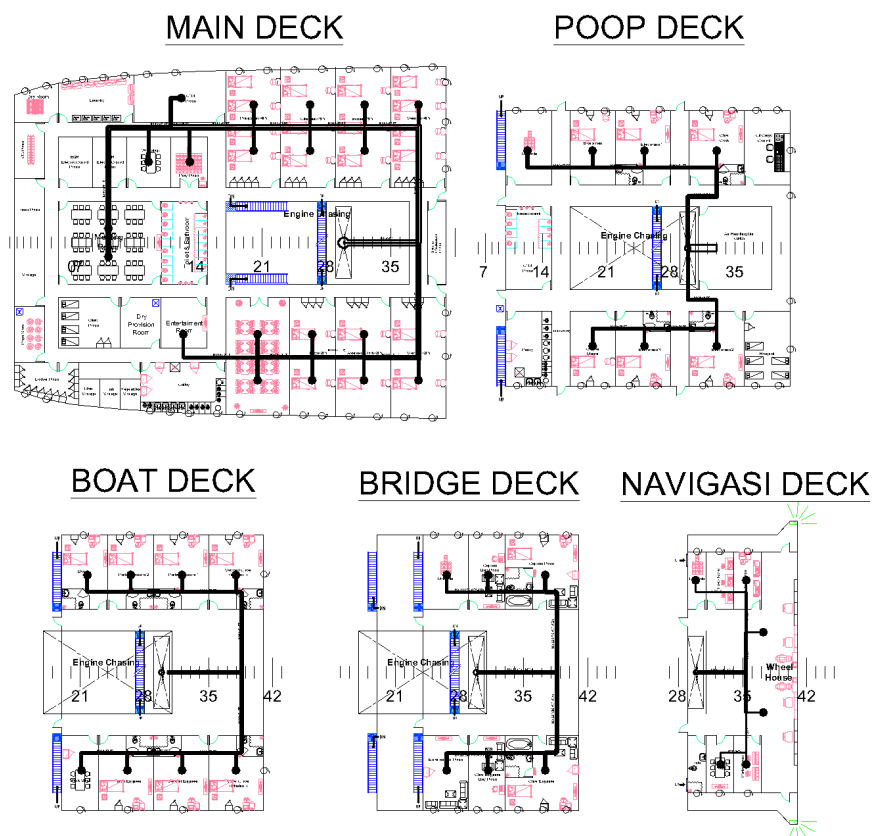
AIR HANDLING UNIT DETAILED DESIGN Calculation Reports & Drawings for MT. ANTARES 1			Prep. :	HMP
			Chk. :	GEK
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- Navigasi Deck



Gambar 5. 9 Duct Routing Bridge Deck

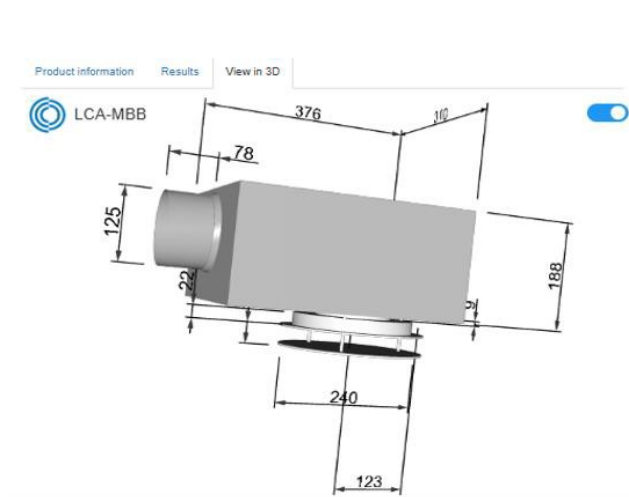
5.8. DUCTWORK DRAWING



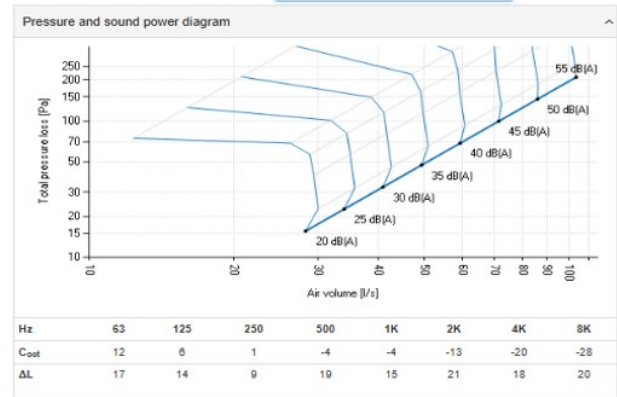
Gambar 5. 10 Ductwork Drawing

AIR HANDLING UNIT DETAILED DESIGN Calculation Reports & Drawings for MT. ANTARES 1			Prep. :	HMP
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5.9. AIR FLOW MODELLING AND SIMULATION



Face velocity	v	13.8	m/s
Total pressure loss	Δp_t	451	Pa
Sound power level	L_{wA}	>00	dB(A)
Sound pressure level	L_{pA}	>00	dB(A)
Throw	$L_{0.2}$	8.7	m



LCA + MB plenum box

LCA + MB		Ød ₁	Ød ₂	B	C	D	e	H*	L	W
		mm	mm	mm	mm	mm	mm	mm	mm	mm
100	100	62	245	78	40	197	237	310	260	
100	125	62	245	78	40	197	237	310	260	
100	160	62	245	78	40	197	237	310	260	
125	125	75	291	78	40	222	262	376	310	
125	160	75	291	78	40	222	262	376	310	
125	200	75	291	78	40	222	262	376	310	
160	160	92	352	78	40	256	296	459	380	
160	200	92	352	78	40	256	296	459	380	
160	250	92	352	78	40	256	296	459	380	
200	200	112	425	78	40	297	337	565	460	
200	250	112	425	78	40	297	337	565	460	
200	315	112	425	78	40	297	337	565	460	
250	250	137	534	118	60	347	387	698	540	
250	315	137	534	118	60	347	387	698	540	
250	400	137	534	118	60	347	387	698	540	
315	315	170	695	118	60	412	452	858	540	
315	400	170	695	118	60	412	452	858	540	

Formo - Ceiling diffuser

LCA

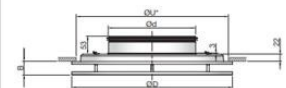


Description

LCA is a circular diffuser with unperforated face plate and can be used for both supply and extract air. LCA is suitable for horizontal supply of cooled air and can be equipped with accessories of various types in order to achieve optimal function.

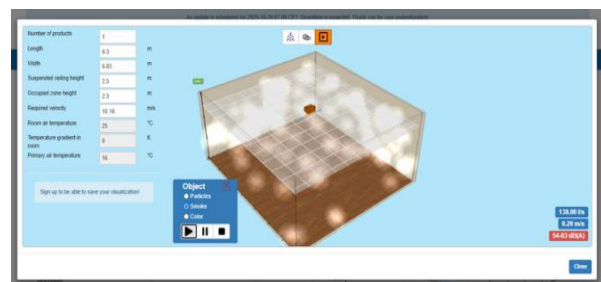
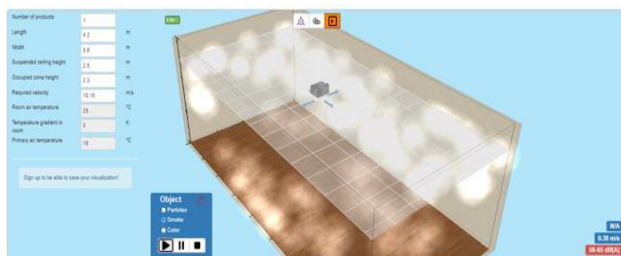
Installing a LCA diffuser in a plenum box type MB or CB can help to achieve a stable airflow to the diffuser as well as realize the potential for individual adjustment.

Dimensions



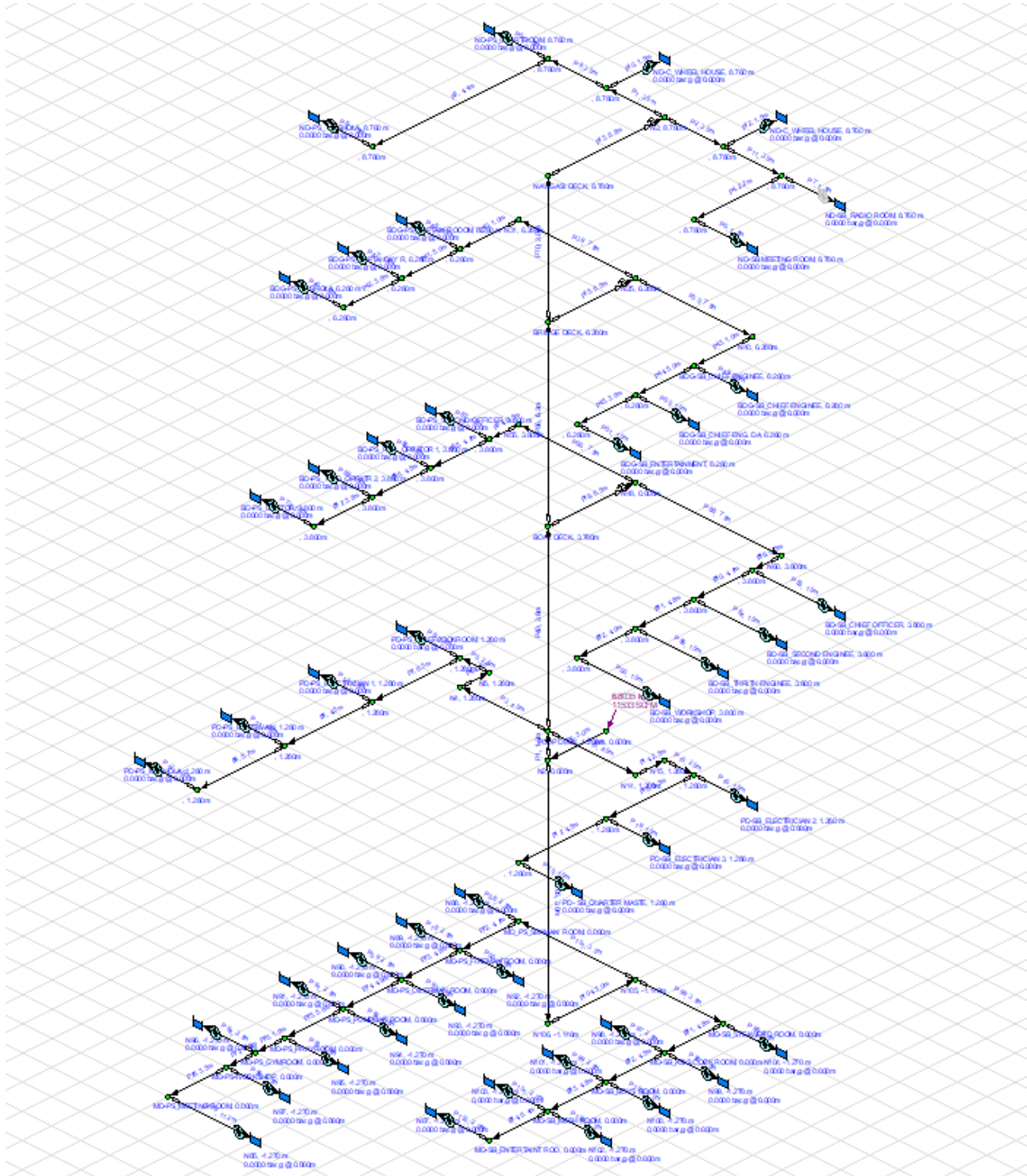
LCA Ød	Ød	Ød1	B	Free area A	m
mm	mm	mm	mm	mm²	kg
100	240	200	37	0.010	0.8
125	240	200	37	0.011	0.9
160	300	260	37	0.0166	1.3
200	360	320	37	0.023	1.6
250	480	420	41	0.030	2.6
315	540	500	41	0.037	3.4
400	540	500	41	0.037	3.3

* ØU = Ceiling grid opening.



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5.10. REFRIGERATION EQUIPMENT LAYOUT AND PLOT PLAN



Gambar 5. 11 Layout Ducting

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dioksida yang terkandung. Berdasarkan tabel perbandingan tiap-tiap refrigerant, maka diketahui bahwa refrigerant type R22 merupakan refrigerant yang paling sesuai digunakan pada sistem ini. Karena memiliki angka GWP yang tidak terlalu tinggi sebesar 1960 mengacu IPPC AR6.

ODP singkatan dari Ozone Depletion Potential, yaitu potensial setiap 1 Kg refrigerant untuk menghancurkan lapisan ozon di atmosfer, tergantung pada setiap 1 Kg Fluorotrichloromethane yang terkandung. Berdasarkan tabel perbandingan tiap-tiap refrigerant, maka diketahui bahwa refrigerant type R22 merupakan refrigerant yang paling sesuai digunakan pada sistem ini. Karena memiliki angka ODP yang tidak tinggi sebesar 0.055, artinya hampir tidak memiliki potensi bahaya ODP.

ISO 817: 2014, mengklasifikasikan refrigerant dalam segi keamanan. ISO 817 menetapkan klasifikasi kemanan refrigerant berdasarkan toxicity (racun) dan flammability (mampu terbakar). Standard ini membagi berdasarkan LFL lower flammable limit, menjadi 3 kelas dengan tambahan sub kelas:

- Class 3 : higher flammability
- Class 2 : lower flammability
- Clas 2L : lower burning velocity (BV) class 2s with burning velocities less than or equal to 10 cm/s
- Class 1 : no flame propagation

Sementara itu klasifikasi toxicity terbagi menjadi 2 kelas berdasarkan paparan:

- Class A : (lower chronic toxicity) refrigerant with an occupational exposure limit of 400ppm or greater
- Class B : (higher chronic toxicity) refrigerant with an occupational exposure limit of less than 400ppm.

Berdasarkan tabel perbandingan tiap-tiap refrigerant, maka diketahui bahwa refrigerant type R22 merupakan refrigerant yang paling sesuai digunakan pada sistem ini. Karena sudah tersertifikasi A1 dari International standard organization (ISO). Secara kelas R22 tidak beracun dan tidak mudah terbakar. Selain itu R22 juga termasuk jenis transitional service refrigerant karena tergolong refrigerant HFC dan chlorin free berdasarkan Structural Classification Of Refrigerants.

AIR HANDLING UNIT DETAILED DESIGN Calculation Reports & Drawings for MT. ANTARES 1			Prep. :	HMP
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5.12. AIR HANDLING UNIT (AHU) SPECIFICATION AND SELECTION

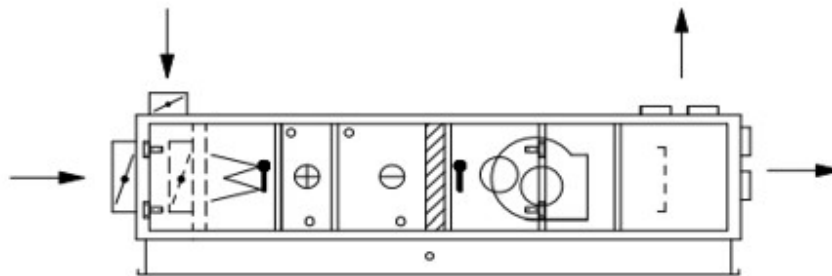
AIR HANDLING UNIT



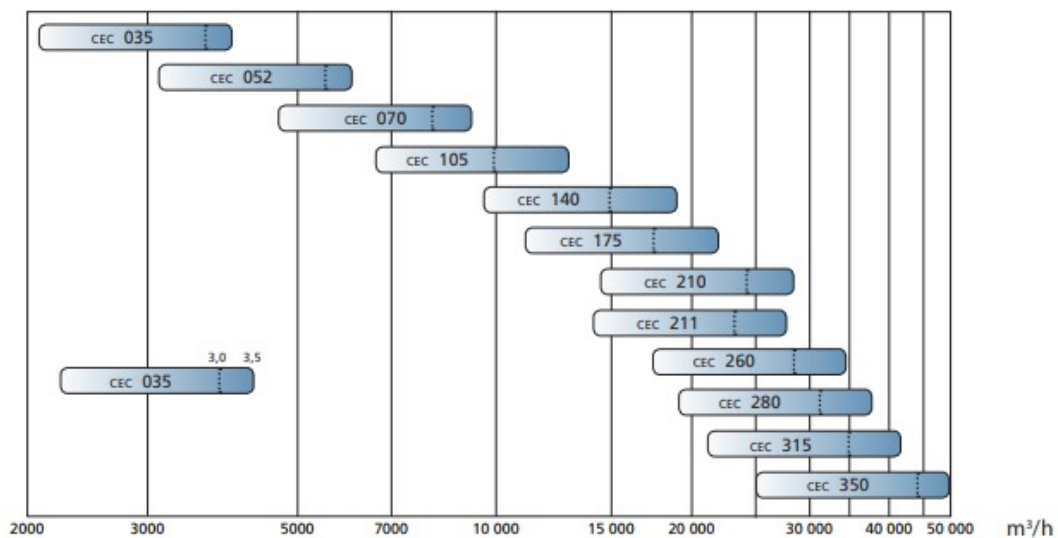
VARIANT 1 – 1-PIPE (SINGLE PIPE) WITH HEATING AND COOLING

The air taken in is 100% fresh air or combined with recycled air in desired proportion, and filtered before undergoing further treatment. The air is then heated or cooled to obtain

the desired temperature on the air leaving the unit. The air is then distributed throughout the ship through Aeron spiro pipe system.



QUICK SELECTION



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GENERAL DIMENSION

				Airflow [m ³ /h]		
Dimension		Filter		Air velocity in filter and cooling coil		
	With [mm]	Hight [mm]	Area [m ²]	2,0 m/s	3,0 m/s	3,5 m/s
CECE 035	740	700	0,36	2400	3600	4200
CECE 052	1050	700	0,54	3500	5300	6200
CECE 070	1050	1010	0,72	5400	8000	9500
CECE 105	1290	1010	1,08	6800	10000	12000
CECE 140	1440	1300	1,44	10000	15000	18000
CECE 175	1730	1300	1,80	12000	18000	21500
CECE 210	1730	1690	2,16	16000	24500	28500
CECE 211	1980	1400	2,16	15000	23000	26500
CECE 260	1980	1690	2,70	19000	28000	33000
CECE 280	2480	1500	2,88	20000	31000	36000
CECE 315	2180	1940	3,24	24000	35000	41000
CECE 350	3120	1690	3,60	29000	44000	51000

5.13. AIRFLOW EVALUATION AND ANALYSIS

- Main Deck
- Poop Deck

STARBOARD								
Sebelum Perubahan			Duct Terminal Size		After Add Diffuser			
CFM Calc	CFM PIPE	PERCENT	Sebelum	Sesudah	CFM Calc	CFM PIPE	PERCENT	
153.00	351	129%	5		153.00		100%	
150.00	165	10%	5		150.00		100%	
153.00	146	-5%	5		153.00		100%	
268.00	141	-47%	0		268.00		100%	
STARBOARD								
Sebelum Perubahan			Duct Terminal Size		After Add Diffuser			
CFM Calc	CFM PIPE	PERCENT	Sebelum	Sesudah	CFM Calc	CFM PIPE	PERCENT	
150.00	194	29%	5		150.00		100%	
199.00	157	-21%	5		199.00		100%	
148.00	143	-3%	5		148.00		100%	

- Boat Deck

PORTSIDE							
Before Change			Duct Terminal Size		After Add Diffuser		
CFM Calc	CFM PIPE	PERCENT	Before	After	CFM Calc	CFM PIPE	PERCENTASE
269.00	270	0%	6		269.00	295	-10%
117.00	107	-9%	4		117.00	109	7%
103.00	97.66	-5%	4		103.00	99.49	3%
213.00	163	-23%	5		213.00	165	23%

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STARBOARD							
Sebelum Perubahan			Duct Terminal Size		After Add Diffuser		
CFM Calc	CFM PIPE	PERCENT	Sebelum	Sesudah	CFM Calc	CFM PIPE	PERCENT
151.00	171	13%	5	4	151.00	171	-13%
151.00	170	13%	5	4	151.00	161	-7%
132.00	150	14%	5	4	132.00	141	-7%
219.00	213	-3%	6	5	219.00	213	3%

- **Bridge Deck**

PORTSIDE							
Before Change			Duct Terminal Size		After Add Diffuser		
CFM Calc	CFM PIPE	PERCENT	Before	After	CFM Calc	CFM PIPE	PERCENTASE
326.00	441	35%	8	5	326.00	384	-18%
151.00	238	58%	6	5	151.00	179	-19%
239.00	264	10%	5	5	239.00	233	3%

STARBOARD							
Sebelum Perubahan			Duct Terminal Size		After Add Diffuser		
CFM Calc	CFM PIPE	PERCENT	Sebelum	Sesudah	CFM Calc	CFM PIPE	PERCENT
428.00	376	-12%	7	7	428.00	410	4%
196.00	240	22%	5	5	196.00	196	0%
335.00	387	16%	6	6	335.00	263	21%

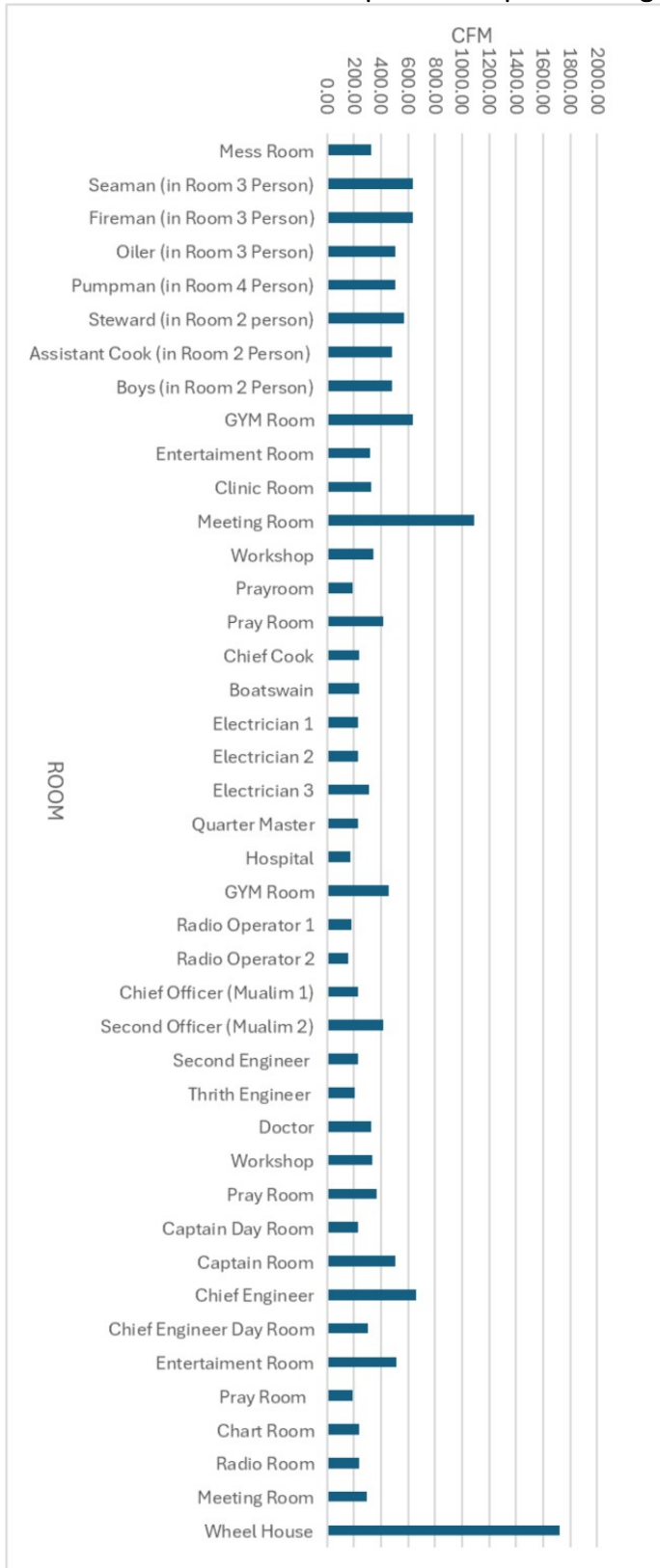
- **Navigasi Deck**

PORTSIDE							
Before Change			Duct Terminal Size		After Add Diffuser		
CFM Calc	CFM PIPE	PERCENT	Before	After	CFM Calc	CFM PIPE	PERCENTASE
558.50	464	-17%	7		558.50		100%
156.00	168	8%	4		156.00		100%
121.00	68.97	-43%	4		121.00		100%

STARBOARD							
Sebelum Perubahan			Duct Terminal Size		After Add Diffuser		
CFM Calc	CFM PIPE	PERCENT	Sebelum	Sesudah	CFM Calc	CFM PIPE	PERCENT
558.50	400	-28%	7		558.50		100%
157.00	501	219%	4		157.00		100%
190.00	141	-26%	4		190.00		100%

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5.13.1. Analisa Kebutuhan Suplai Udara pada Ruangan



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Data hasil perhitungan beban panas,laju aliran udara,dan dimensi ducting dari setiap ruangan yang membutuhkan pengkondisian udara disajikan pada tabel berikut :

Tabel. Hasil Summary calculation AHU tiap Deck

SPACE	DECK	SIDE	FRAMES	Volume (ft ³)	Heat Area (ft ²)	BTU/HR			SHF	OUTSIDE AIR BASED ON			CFM	R/C TOTAL AIR	Air Quantity (l/s)
						SENS.	LATENT	TOTAL		R/C	PERS.	EXH.			
ss Room	Main Deck	P	17 to 24	2088.16	1295.50	5718.29	2000	7718.285	0.741	69.60527	60	60	755.00	69.60527	356.36
man (in Room 3 Person)		P	35 to 41	4449.65	2456.32	11111.32	1500	12611.32	0.88106	148.3217	60	60	412	148.3217	194.464
man (in Room 3 Person)		P	29 to 35	4449.65	3129.82	11104.43	1500	12604.43	0.88099	148.3217	45	45	412	148.3217	194.464
er (in Room 3 Person)		P	23 to 29	4449.65	3134.99	8904.24	1500	10404.24	0.856	148.3217	45	45	330.00	148.3217	155.76
Pumpman (in Room 4 Person)		P	17 to 23	4449.65	2390.93	8881.95	1500	10381.95	0.856	148.3217	72	72	329.00	148.3217	155.29
ward (in Room 2 person)		S	35 to 41	4449.65	2454.26	9940.67	1000	10940.67	0.909	148.3217	30	30	369.00	148.3217	174.17
Assistant Cook (in Room 2 Person)		S	29 to 35	4449.65	2450.64	8429.16	1000	9429.163	0.894	148.3217	30	30	313.00	148.3217	147.74
M Room		S	24 to 29	4449.65	2450.64	8429.16	1000	9429.163	0.894	148.3217	30	30	270.00	148.3217	127.44
entertainment Room		S	6 to 17	3150.07	1831.84	11078.53	3000	14078.53	0.787	105.0024	90	90	411.00	105.0024	193.99
ic Room		S	-4 to 5	1546.78	1022.14	5535.08	2000	7535.083	0.735	51.55946	60	60	206.00	60	97.23
eting Room		S	-4 to 5	2088.16	1295.50	5718.29	2000	7718.285	0.741	69.60527	60	60	212.00	69.60527	100.06
rkshop		C	(-5) to 10	5728.93	2274.73	19072.67	10000	29072.67	0.656	190.9642	300	300	707.00	300	333.70
lyroom		P	5 to 11	1476.15	981.24	5930.97	2000	7930.967	0.748	49.20515	90	90	220.00	90	103.84
Pray Room		P	11 to 15	1243.08	864.04	3243.53	2000	5243.533	0.619	41.43591	60	60	186.00	60	87.79
Chief Cook	Poop Deck	P	10 to 15	2295.46	1410.07	7230.83	2000	9230.827	0.783	76.51518	60	60	268.00	76.51518	126.50
Boatswain		P	30 to 36	2295.46	876.72	4122.64	500	4622.64	0.89184	76.51518	15	15.00	153	76.51518	72.216
Electrician 1		P	17 to 22	2754.55	1565.07	4116.01	500	4616.011	0.89168	91.81822	15	15	153	91.81822	72.22
Electrician 2		P	22 to 28	2754.55	1621.04	4040.29	500	4540.285	0.890	91.81822	15	15	150.00	91.81822	70.80
Electrician 3		S	30 to 36	2754.55	1621.04	4040.29	500	4540.285	0.890	91.81822	15	15	150.00	91.81822	70.80
Quarter Master		S	22 to 28	2754.55	1621.04	5366.06	1000	6366.055	0.843	91.81822	15	15	199.00	91.81822	93.93
spital		S	17 to 22	2295.46	1410.07	3972.63	1000	4972.626	0.799	76.51518	15	15	148.00	76.51518	69.86
GYM Room		S	36 to 41	1271.33	879.41	3023.34	1000	4023.341	0.751	42.37764	15	15	112.00	42.37764	52.86
Radio Operator 1		C	10 to 15	1271.33	876.72	8017.48	2000	10017.48	0.800	42.37764	15	15	297.00	42.37764	140.18
Radio Operator 2	Boat Deck	P	29 to 35	2754.55	1562.38	3132.11	500	3632.11	0.862	91.81822	15	15	117.00	91.81822	55.22
Chief Officer (Mualim 1)		P	24 to 29	2295.46	1407.38	2777.89	500	3277.892	0.84746	76.51518	15	15	103	76.51518	48.62
Second Officer (Mualim 2)		S	35 to 41	2754.55	1618.35	4072.61	500	4572.612	0.89065	91.81822	15	15	151	91.81822	71.27
Thrith Engineer		P	35 to 41	2754.55	1618.35	7262.19	500	7762.193	0.93559	91.81822	15	15	269	91.81822	126.97
ctor		S	29 to 35	2754.55	1618.35	4072.61	500	4572.612	0.891	91.81822	15	15	151.00	91.81822	71.27
Workshop		S	23 to 29	2754.55	1618.35	3544.67	500	4044.67	0.876	91.81822	15	15	132.00	91.81822	62.30
Pray Room		P	19 to 24	2295.46	1407.38	5748.08	500	6248.078	0.920	76.51518	15	15	213.00	76.51518	100.54
Captain Day Room	Bridge Deck	S	19 to 23	1836.36	1196.41	5894.54	2000	7894.536	0.747	61.21215	60	60	219.00	61.21215	103.37
Captain Room		P	24 to 29	2295.46	1407.38	6427.32	1000	7427.317	0.865	76.51518	30	30	239.00	76.51518	112.81
Chief Engineer		P	29 to 33	1836.36	1248.07	4053.05	500	4553.048	0.89018	61.21215	15	15	151	61.21215	71.27
Chief Engineer Day Room		P	33 to 41	2901.94	2008.95	8798.09	500	9298.085	0.946	96.73138	15	15	326.00	96.73138	153.87
Entertainment Room		S	33 to 41	3672.73	2329.85	11536.02	500	12036.02	0.958	122.4243	15	15	428.00	122.4243	202.02
Pray Room	Navigation Deck	S	29 to 33	1836.36	1196.41	5270.94	500	5770.94	0.913	61.21215	15	15	196.00	61.21215	92.51
Chart Room		S	24 to 29	2295.46	1235.16	9018.83	2000	11018.83	0.818	76.51518	15	15	335.00	76.51518	158.12
Radio Room		P	29 to 31	706.29	618.97	3253.06	1000	4253.057	0.765	23.54313	30	30	121.00	30	57.11
Meeting Room		P	34 to 37	1059.44	791.19	4191.47	1000	5191.468	0.807	35.3147	30	30	156.00	35.3147	73.63
Wheel House		S	34 to 37	1059.44	791.19	4213.43	1000	5213.433	0.808	35.3147	30	30	157.00	35.3147	74.10
		S	31 to 34	1059.44	1006.47	5113.99	2000	7113.995	0.719	35.3147	60	60	190.00	60	89.68
		C	29 to 41	12148.27	5058.02	30151.91	2000	32151.91	0.938	404.9422	75	75	1117.00	404.9422	527.22
															0
						295560.6444	58500	354060.6		11533.00					

Tabel 1. Size Main Ducting Yang Didistribusikan

Room	Air Out		
Main Deck	5132.00		
Poop Deck	1630.00		
Boat Deck	1355.00		
Bridge Deck	1675.00		
Navigation Deck	1741.00		
Total	11533.00	CFM =	5443.576 l/s

Gambar 2. 16

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5.14. Results and Discussion

6. CONCLUSIONS AND RECOMMENDATIONS

6.1. Conclusions

Tuliskan kesimpulan di sini untuk menjelaskan hasil pekerjaan **mengacu** dan **menjawab** pada scope of work yang sudah dituliskan. Jumlah Kesimpulan mengacu jumlah scope of work yg sudah disusun :

1.
2.
3.

6.2. Recommendations

Tuliskan rekomendasi-rekomendasi untuk penyempurnaan hasil pekerjaan yg mana rekomendasi itu belum dilakukan atau sebaiknya dilakukan untuk menambah kualitas pekerjaan.

7. ATTACHMENT

Cantumkan Lampiran data, gambar-gambar, simulasi dll yang mendukung pada proses pencapaian pekerjaan di scope of work.

Lampiran 1 Gambar Desain General Arrangements (GA)

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